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Electronic apparatus and accessory

Abstract

An electronic apparatus to which an accessory is to be detachably attached includes a plurality of contacts electrically connectable to the accessory and arranged in a first direction orthogonal to an attachment direction of the accessory. The plurality of contacts include an attachment detection contact that is used to detect an attachment of the accessory to the electronic apparatus, and a first clock contact that is used to communicate a first clock signal in communication between the electronic apparatus and the accessory. A potential of the attachment detection contact changes from a predetermined potential to a reference potential different from the predetermined potential after the accessory is attached to the electronic apparatus. The attachment detection contact is disposed next to the first clock contact.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of International Patent Application No. PCT/JP2021/014282, filed on Apr. 2, 2021, which claims the benefit of Japanese Patent Applications Nos. 2020-070625, filed on Apr. 9, 2020, 2020-070626, filed on Apr. 9, 2020, 2020-070627, filed on Apr. 9, 2020, and 2021-060417, filed on Mar. 31, 2021, each of which is hereby incorporated by reference herein in their entirety.

BACKGROUND

Technical Field

(1) The disclosure relates to an electronic apparatus and an accessory, each of which has a contact that is used for communication, power supply, and the like.

Description of the Related Art

(2) An accessory, such as a strobe unit, is attached to an accessory shoe provided to an electronic apparatus, such as a camera. The accessory shoe includes a plurality of contacts (terminals) for supplying power to the accessory and for communicating with the accessory. In communication between the electronic apparatus and the accessory, signals such as a clock signal and data signal are transmitted from the electronic apparatus via the terminals of the accessory shoe.

(3) However, if the contact arrangement pitch is made narrower so as to increase the number of contacts without increasing the sizes of the accessory shoe and accessory, changes of potentials of the clock signal and data signal will be likely to affect signals connected to contacts adjacent to the contacts of these signals.

(4) Japanese Patent Laid-Open No. ("JP") 2013-34172 discloses a contact arrangement configured to prevent other signals from being affected by noises caused by the clock signal between the electronic apparatus and the accessory, by disposing contacts having a GND potential as a reference potential on both sides of the clock signal contact.

(5) However, in the contact arrangement disclosed in JP 2013-34172, the GND contacts are arranged on both sides of the clock signal contact, the number of contacts increases and the accessory shoe becomes large.

SUMMARY

(6) The disclosure provides an electronic apparatus and an accessory, each of which can reduce the noise influence of a clock signal on a signal of an adjacent contact while suppressing an increase in the number of contacts.

(7) An electronic apparatus according to one aspect of the disclosure to which an accessory is to be detachably attached includes a plurality of contacts electrically connectable to the accessory and arranged in a first direction orthogonal to an attachment direction of the accessory. The plurality of contacts include an attachment detection contact that is used to detect an attachment of the accessory to the electronic apparatus, and a first clock contact that is used to communicate a first clock signal in communication between the electronic apparatus and the accessory. A potential of the attachment detection contact changes from a predetermined potential to a reference potential different from the predetermined potential after the accessory is attached to the electronic apparatus. The attachment detection contact is disposed next to the first clock contact.

(8) An accessory according to another aspect of the disclosure detachably attached to an electronic apparatus includes a plurality of contacts electrically connectable to the electronic apparatus and arranged in a first direction orthogonal to an attachment direction to the electronic apparatus. The plurality of contacts include an attachment detection contact that is used to detect an attachment of the accessory to the electronic apparatus, and a first clock contact that is used to communicate a

first clock signal in communication between the electronic apparatus and the accessory. The attachment detection contact is connected to a reference potential. The attachment detection contact is disposed next to the first clock contact.

(9) Further features of the disclosure will become apparent from the following description of exemplary embodiments (with reference to the attached drawings).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram illustrating configurations of a camera and an accessory according to one embodiment of the disclosure.
- (2) FIGS. 2A to 2C illustrate an example of the camera to which the accessory is attached and their contact arrangements according to the embodiment.
- (3) FIGS. 3A and 3B illustrate how an external force is applied to the accessory attached to the camera according to the embodiment.
- (4) FIG. 4A is a diagram illustrating a configuration for determining a connection state of a grounded contact according to the embodiment.
- (5) FIG. 4B is a flowchart showing processing executed by the camera according to the embodiment.
- (6) FIGS. 5A and 5B show a flowchart of processing executed by the camera according to the embodiment.
- (7) FIGS. 6A, 6B, and 6C show timing charts illustrating a signal change when a contact adjacent to a power supply contact is short-circuited according to the embodiment.
- (8) FIG. 7 is a diagram illustrating an assignment example of a functional signal to a type of the accessory according to the embodiment.
- (9) FIGS. 8A, 8B, and 8C illustrate a connection destination configuration of the functional signal according to the embodiment.
- (10) FIG. 9 is a diagram illustrating a configuration example of the camera and the accessory according to the embodiment.
- (11) FIG. 10 is a diagram illustrating another configuration example of the camera and accessories according to the embodiment.
- (12) FIG. 11 is a diagram illustrating still another configuration example of the camera and accessories according to the embodiment.
- (13) FIG. 12 is a diagram illustrating a configuration example of the accessory according to the embodiment.
- (14) FIG. 13 is a diagram illustrating a configuration example of the camera, the accessory, and an intermediate accessory according to the embodiment.
- (15) FIG. 14 illustrates another configuration example of the camera, the accessory, and the intermediate accessory according to the embodiment.
- (16) FIGS. 15A and 15B are timing charts when the accessory is a strobe device according to the embodiment.
- (17) FIGS. 16A, 16B, and 16C are perspective views of the camera and an external flash unit as the accessory according to an embodiment.
- (18) FIGS. 17A and 17B are an exploded view and a perspective view of an accessory shoe according to the embodiment.
- (19) FIGS. 18A, 18B, and 18C illustrate a structure of an engagement member and a connection terminal connector of an accessory shoe according to the embodiment.
- (20) FIGS. 19A and 19B are a perspective view and a sectional view of the external flash unit according to the embodiment.

- (21) FIGS. 20A, 20B, and 20C are a perspective view and a front view illustrating an internal structure of a camera connector according to the embodiment.
- (22) FIGS. 21A and 21B are a top view and a sectional view of a camera connector according to the embodiment.
- (23) FIGS. 22A and 22B are a perspective view and a sectional view of an external flash unit according to a variation.
- (24) FIGS. 23A and 23B are a perspective view and a front view illustrating the internal structure of the connector according to the variation.
- (25) FIG. 24 is a front view of the accessory shoe according to the embodiment.
- (26) FIG. 25 is a partially enlarged view of a connection plug according to the embodiment.
- (27) FIG. 26 is a front sectional view illustrating a state in which a camera connector is attached to an accessory shoe according to the embodiment.

DESCRIPTION OF THE EMBODIMENTS

- (28) Referring now to the accompanying drawings, a description will be given of embodiments according to the disclosure.
- (29) FIG. 1 illustrates an electrical configuration of a camera **100** as an electronic apparatus according to one embodiment of the disclosure and an accessory **200** detachably attached to the camera **100**. In the camera **100** and the accessory **200**, a plurality of contacts (terminals) TC01 to TC21 of a camera connector **141** provided in the camera **100** and a plurality of contacts TA01 to TA21 of an accessory connector **211** provided in the accessory **200** are connected to each other in a one-to-one correspondence, respectively and thereby they are electrically connected with each other.
- (30) The camera **100** is supplied with the electric power from a battery **111**. The battery **111** is attachable to and detachable from the camera **100**. A camera control circuit **101** as a control means of the camera **100** is a circuit that controls the entire camera **100**, and includes a microcomputer having a built-in CPU or the like.
- (31) A system power supply circuit **112** is a circuit that generates a power supply for supplying the power to each circuit in the camera **100**, and includes a DC/DC converter circuit, an LDO (Low Drop Out), a charge pump circuit, and the like. A voltage of 1.8V generated by the system power supply circuit **112** is constantly supplied as a camera microcomputer power supply VMCU_C from the battery **111** to the camera control circuit **101**. The camera control circuit **101** controls the system power supply circuit **112** and thereby controls turning on and off of the power supply to each circuit of the camera **100**.
- (32) An optical lens **121** is attachable to and detachable from the camera **100**. Light from an object incident through the optical lens **121** is imaged on an image sensor **122** including a CMOS sensor, a CCD sensor, or the like. The object image formed on the image sensor **122** is encoded into a digital imaging signal. An image processing circuit **123** performs image processing such as noise reduction processing and white balance processing for the digital imaging signal to generate image data, and converts it into an image file, such as a JPEG format, in order to record the image data in a recording memory **126**. The image processing circuit **123** generates from the image data VRAM image data for display on a display circuit **127**.
- (33) A memory control circuit **124** controls a transmission and reception of the image data and other data generated by the image processing circuit **123** and the like. A volatile memory **125** is a memory capable of high-speed reading and writing such as DDR3 SDRAM, and is used for a workspace or the like of the image processing performed by the image processing circuit **123**. The recording memory **126** is a readable and writable recording medium such as an SD card or CFexpress card that is attachable to and detachable from the camera **100** via an unillustrated connector. The display circuit **127** is a display located on the back surface of the camera **100**, and includes an LCD panel, an organic EL display panel, and the like. A backlight circuit **128** adjusts the brightness of the display circuit **127** by changing a light amount of the backlight of the display

circuit **127**.

(34) Each of an accessory-use power supply circuit **A131** and an accessory-use power supply circuit **B132** as the power supply means is a voltage conversion circuit that converts the voltage supplied from the system power supply circuit **112** into a predetermined voltage, and generates 3.3 V as an accessory power supply VACC in this embodiment.

(35) The accessory-use power supply circuit **A131** is a power supply circuit including an LDO or the like and having a low self-consumption power. The accessory-use power supply circuit **B132** is a circuit that includes a DC/DC converter circuit or the like and can flow a larger current than that of the accessory-use power supply circuit **A131**. The self-consumption power of the accessory-use power supply circuit **B132** is larger than that of the accessory-use power supply circuit **A131**. Therefore, when the load current is small, the accessory-use power supply circuit **A131** is more efficient than the accessory-use power supply circuit **B132**, and when the load current is large, the accessory-use power supply circuit **B132** is more efficient than the accessory-use power supply circuit **A131**. The camera control circuit **101** controls turning on and off of the voltage outputs of the accessory-use power supply circuits **A131** and **B132** according to the operating state of the accessory **200**.

(36) A protection circuit **133** as a protection means includes a current fuse element, a poly-switching element, an electronic fuse circuit that combines a resistor, an amplifier, and a switching element, and the like, and outputs an overcurrent detection signal DET_OVC when the power supply current values from the accessory-use power supply circuits **A131** and **B132** to the accessory **200** become excessive (abnormal) beyond a predetermined value. In this embodiment, the protection circuit **133** includes an electronic fuse circuit, and notifies the camera control circuit **101** through the overcurrent detection signal DET_OVC when the current of 1 A or higher flows. The overcurrent detection signal DET_OVC indicates the overcurrent through the Hi level.

(37) A camera connector **141** is a connector for an electrical connection with the accessory **200** via 21 contacts TC01 to TC21 arranged in a row. The contacts TC01 to TC21 are arranged in this order from one end to the other end in the arrangement direction thereof.

(38) TC01 is connected to the ground (GND) and serves not only as a contact for the reference potential (GND potential) but also as a contact for controlling the wiring impedance of differential signals D1N and D1P described below. TC01 corresponds to a third grounded contact.

(39) The differential signal MN connected to TC02 and the differential signal D1P connected to TC03 are paired differential data communication signals that perform data communication, and connected to the camera control circuit **101**. TC02, TC03, TC07 to TC17, TC19, and TC20, which will be described later, are communication contacts.

(40) TC04 as a first grounded contact is connected to GND and serves as a reference potential contact between the camera **100** and the accessory **200**. TC04 is disposed outside TC05 described below in the contact arrangement direction.

(41) The accessory power supply VACC generated by the accessory-use power supply circuits **A131** and **B132** is connected to TC05 as the power supply contact via the protection circuit **133**.

(42) An accessory attachment detection signal/ACC_DET is connected to TC06 as an attachment detection contact. The accessory attachment detection signal/ACC_DET is pulled up to the camera microcomputer power supply VMCU_C via a resistor element Rp134 (10 kΩ). The camera control circuit **101** can detect whether or not the accessory **200** is attached, by reading the signal level of the accessory attachment detection signal/ACC_DET. If the accessory attachment detection signal/ACC_DET signal level (potential) is a Hi level (predetermined potential), the accessory **200** is detected as being non-attached, and if it is a Lo level (GND potential as described later), the accessory **200** is detected as being attached.

(43) When the camera **100** is powered on and the signal level (potential) of the accessory attachment detection signal/ACC_DET changes from the Hi level to the Lo level, various transmissions are performed between the camera **100** and the accessory **200** via the contacts.

- (44) The camera control circuit **101** supplies the power to the accessory **200** via TC**05** as a power supply contact when the attachment state of the accessory **200** is detected.
- (45) SCLK connected to TC**07**, MOSI connected to TC**08**, MISO connected to TC**09**, and CS connected to TC**10** are signals used by the camera control circuit **101** as a communication master to perform a SPI (Serial Peripheral Interface) communication. In this embodiment, the communication clock frequency of the SPI communication is 1 MHz.
- (46) A communication request signal/WAKE for requesting a communication from the accessory **200** to the camera control circuit **101** is connected to TC**11**. The communication request signal/WAKE is pulled up to the camera microcomputer power supply VMCU_C via a resistor. The camera control circuit **101** can receive the communication request from the accessory **200** by detecting a trailing edge of the communication request signal/WAKE.
- (47) SDA connected to TC**12** and SCL connected to the TC**13** are signals for the camera control circuit **101** to act as a communication master and perform an I2C (Inter-Integrated Circuit) communication. SDA and SCL are signals for an open drain communication (referred to as an open drain communication hereinafter) pulled up to the camera microcomputer power supply VMCU_C, and the communication frequency is 100 kbps in this embodiment.
- (48) In the I2C communication, both the data transmission from the camera **100** and the data transmission from the accessory **200** are performed via the SDA. When the SPI communication and the I2C communication are compared with each other, the I2C communication has a lower communication speed than that of the SPI communication, and can achieve a lower power consumption. The SPI communication has a higher communication speed than the I2C communication, and is therefore suitable for a communication of information having a large data amount. Therefore, in the communication between the camera **100** and the accessory **200** according to this embodiment, information having a large data amount is communicated in the SPI communication, and information having a small data amount is communicated in the I2C communication. For example, data is first communicated in the I2C communication, and control is made so that the SPI communication is further executed when the SPI communication can be executed or the SPI communication needs to be executed based on this data.
- (49) An FNC**1** signal connected to TC**14** (synchronous contact), an FNC**2** signal connected to TC**15**, an FNC**3** signal connected to TC**16**, and an FNC**4** signal connected to TC**17** are signals that can change the function according to the type of the attached accessory **200**. For example, when the accessory **200** is a microphone device, the signal communicated via TC **15** is an audio data signal. When the accessory **200** is an illumination (strobe or flash) unit, the signal communicated via TC**14** is a signal for controlling the light emission timing. Depending on the type of the attached accessory, a signal that realizes a different function may be communicated via the same contact. For example, when the accessory **200** is an accessory other than the illumination unit, a synchronization signal for controlling a timing different from the light emission timing may be communicated via TC**14**. TC**14** to TC**17** correspond to functional signal contacts. Communication using at least one of the functional signal contacts is also referred to as a functional signal communication.
- (50) The functional signal communication can execute communication at a timing that does not depend on the I2C communication or the SPI communication in parallel with the I2C communication and the SPI communication.
- (51) The type of the accessory, as used herein, means the above-mentioned microphone device, illumination unit, and the like. Accessories that achieve the same purpose, such as illuminations with different performances, belong to the same type. Accessories that achieve different purposes, such as a microphone device and an illumination unit, belong to different types.
- (52) The functional signal communication is executed based on the information acquired by the I2C communication or the SPI communication.
- (53) TC**18** as a second grounded contact (reference potential contact) is also connected to GND,

and is a contact that serves as a reference potential between the camera **100** and the accessory **200**, similar to TC**04**.

(54) A differential signal D2N connected to TC**19** (first differential signal contact) and a differential signal D2P connected to TC**20** (second differential signal contact) are paired data communication signals that perform data communication and connected to the camera control circuit **101**. For example, the USB communication can be performed via TC**19** and TC**20**.

(55) TC**21** is connected to GND and can be used not only as a contact for a reference potential but also as a contact for controlling the wiring impedance of the differential signals D2N and D2P. TC**21** corresponds to a fourth grounded contact. The contacts TC**01**, TC**04**, TC**06**, TC**18**, and TC**21** are connected, for example, to a GND portion of a flexible substrate **158** illustrated in FIGS. **17A** and **17B**, which will be described later, and the GND portion of the flexible substrate **158** is fixed with a metallic member having a GND level of the camera **100** by a screw **157** or the like. The metallic member having the GND level includes, for example, an engagement member **151**, an unillustrated base plate inside the camera **100**, and the like.

(56) This embodiment disposes the attachment detection contact TC**06** to which the accessory attachment detection signal/ACC_DET is connected is arranged, next to the contact (first clock contact) TC**07** that transmits SCLK (first clock signal) as a clock signal. In general, a noise (clock noise) due to the potential fluctuation of the clock signal is transmitted to a contact adjacent to the contact of the clock signal, which may cause a malfunction. In particular, in a configuration having a large number of contacts and a short distance between contacts as in this embodiment, the influence is more significant. Accordingly, disposing the attachment detection contact TC**06** next to the SCLK contact TC**07** can suppress the influence of the clock noise.

(57) The accessory attachment detection signal/ACC_DET is pulled up before the accessory is attached, but is set to the GND potential after the accessory is attached. On the other hand, the SCLK contact TC**07** that transmits the clock signal does not transmit the clock signal before the accessory is attached, and thus the potential does not fluctuate. The potential fluctuates because the clock signal is transmitted only after the accessory is attached.

(58) When the SCLK contact TC**07** transmits the clock signal, the attachment detection contact TC**06** is at the GND potential. Therefore, even if the attachment detection contact TC**06** receives the clock noises, the potential of the control circuit of the camera **100** or the accessory **200** is less likely to fluctuate, so that the malfunction can be prevented. In addition, the clock noise can be restrained from transmitting to a position distant farther than the attachment detection contact TC**06**. As a result, since it is unnecessary to dispose the GND terminal, the influence of the clock noises can be suppressed without increasing the number of contacts.

(59) SCL (second clock signal) as a clock signal is also transmitted to the contact (second clock contact) TC**13**. However, the frequency of SCLK transmitted to the SCLK contact TC**07** is higher than that of SCL, and the SCLK contact TC**07** generates more clock noises than the SCL contact TC**13**. Therefore, disposing the attachment detection contact TC**06** next to the SCLK contact TC**07** rather than next to the SCL contact TC**13** can provide a greater effect of preventing the malfunction due to the clock noises.

(60) In addition to the difference in frequency, SCL transmitted by the SCL contact TC**13** is a clock signal of the I2C communication standard, and the voltage fluctuation of the signal line is driven by the open drain connection. On the other hand, SCLK transmitted by the SCLK contact TC**07** is a clock signal pursuant to the SPI communication standard, and the voltage fluctuation of the signal line is driven by the CMOS output. Therefore, the SCL contact TC**13** tends to have a gentler edge of the voltage fluctuation than the SCLK contact TC**07**, and the clock noises are less likely to occur. Thus, disposing the attachment detection contact TC**06** next to the SCLK contact TC**07** rather than next to the SCL contact TC**13** is more effective in preventing the malfunction caused by the clock noises.

(61) The differential signals D1N and D1P may be transmitted in pairs to the first and second

differential signal contacts TC19 and TC20 to transmit the clock signal. At that time, a clock signal (third clock signal) having a frequency higher than that of the SCLK contact TC07 or the SCL contact TC13 may be transmitted. Since the differential signals D1N and D1P are paired signals, the emission of the clock noise is smaller than that of the SCLK contact TC07 or the SCL contact TC13 that transmits a single-ended signal. Therefore, disposing the attachment detection contact TC06 next to the SCLK contact TC07 rather than next to the first and second differential signal contacts TC19 and TC20 can more effectively prevent the malfunction caused by the clock noise.

(62) The contact (first data contact) TC08 disposed on an opposite side of the attachment detection contact TC06 with respect to the SCLK contact TC07 transmits MOSI (first data signal). Since MOSI is a data signal, it appears to be susceptible to clock noises. Since MOSI is a data signal of the same SPI communication standard as the clock signal transmitted by the SCLK contact TC07, the fluctuation timing of the potential is synchronized with the clock signal and is less affected by the clock noise. Therefore, the contact TC08 does not have to be fixed to the GND potential and can be used as a MOSI contact.

(63) The accessory 200 has a battery 205 and receives the power supply from the battery 205 and also receives the power supply from the camera 100 via the camera connector 141 and the accessory connector 211. An accessory control circuit 201 as a control means of the accessory 200 is a circuit that controls the entire accessory 200, and is a microcomputer having a built-in CPU and the like.

(64) An accessory power supply circuit 202 is a circuit that generates a power supply to be supplied to each circuit of the accessory 200, and includes a DC/DC converter circuit, an LDO, a charge pump circuit, and the like. The voltage 1.8 V generated by the accessory power supply circuit 202 is constantly supplied as an accessory microcomputer power supply VMCU_A to the accessory control circuit 201. Turning on and off of the power supply to each circuit of the accessory 200 is controlled by controlling the accessory power supply circuit 202.

(65) A charging circuit 204 is a circuit for charging the battery 205 using the power supplied from the camera 100. When it can be determined that the sufficient power is supplied from the camera 100 to perform the charging operation, the accessory control circuit 201 controls the charging circuit 204 to charge the battery 205. Although the battery 205 attached to the accessory 200 has been described in this embodiment, the accessory 200 may operate only with the power supply from the camera 100 without the battery 205. In this case, no charging circuit 204 is necessary.

(66) A differential communication circuit 207 is a circuit for performing a differential communication with the camera 100, and can transmit and receive data to and from the camera 100. An external communication IF circuit 208 is an IF circuit for performing data communication with an unillustrated external device, such as an Ethernet communication IF, a wireless LAN communication IF, and a public network communication IF.

(67) The accessory control circuit 201 controls the differential communication circuit 207 and the external communication IF circuit 208 and thereby can transmit the data received from the camera 100 to the external device or the data received from the external device to the camera 100. A functional circuit 206 is a circuit having a different function depending on the type of the accessory 200. A configuration example of the functional circuit 206 will be described later.

(68) An external connection terminal 209 is a connector terminal connectable to an external device, and is a USB TYPE-C connector in this embodiment. A connection detecting circuit 210 is a circuit for detecting that an external device is connected to the external connection terminal 209, and the accessory control circuit 201 can detect that the external device has been connected to the external connection terminal 209 by receiving the output signal of the connection detecting circuit 210.

(69) A power switch 203 is a switch for turning on and off the operation of the accessory 200, and the accessory control circuit 201 can detect the ON position and the OFF position by reading the signal level of the terminal to which the power switch 203 is connected.

(70) The accessory connector 211 is a connector electrically connectable to the camera 100 via the

21 contacts TA01 to TA21 that are arranged in a row. The contacts TA01 to TA21 are arranged in this order from one end to the other end in the arrangement direction.

(71) TA01 is connected to GND and serves not only as a contact of the reference potential but also as a contact for controlling the wiring impedance of the differential signals DIN and D1P. TA01 corresponds to a third grounded contact.

(72) The differential signal DIN connected to TA02 and the differential signal D1P connected to TA03 are paired data communication signals for the data communication, and are connected to the differential communication circuit 207. TA02, TA03, TA07 to TA17, TA19, and TA20 described later are communication contacts.

(73) TA04 as the first grounded contact is connected to GND and serves as a reference potential contact between the camera 100 and the accessory 200. TA04 is located outside TA05 described below in the contact arrangement direction.

(74) The accessory power supply circuit 202 and the charging circuit 204 are connected to TA05 as the power supply contact, and the accessory power supply VACC supplied from the camera 100 is connected to it.

(75) TA06 as an attachment detection contact is directly connected to GND, and turns the accessory attachment detection signal/ACC_DET described above into the GND level as the Lo level when the accessory 200 is attached to the camera 100. Thereby, it becomes it serves as a contact for causing the camera 100 to detect the attachment of the accessory 200.

(76) SCLK connected to TA07, MOSI connected to TA08, MISO connected to TA09, and CS connected to TA10 are signals for the accessory control circuit 201 to act as a communication slave and perform the SPI communication.

(77) The communication request signal/WAKE for requesting the communication from the accessory control circuit 201 to the camera 100 is connected to TAW. When the accessory control circuit 201 outputs the communication request signal/WAKE at the Lo level and requests the camera 100 for the communication when determining that the communication with the camera 100 is necessary.

(78) When the power is supplied from the camera control circuit 101 to the accessory 200 via TC5 in response to detecting that the accessory 200 is in the attached state, the accessory control circuit 201 notifies the camera control circuit 101 of the reception of the power supply by changing the signal level (potential) of the communication request signal/WAKE from the Hi level to the Lo level.

(79) The accessory control circuit 201 notifies that the accessory 200 has a cause of the communication with the camera 100 by changing the signal level (potential) of the communication request signal/WAKE from the Hi level to the Lo level even without any requests from the camera. With this configuration, the camera control circuit 101 can omit the operation of periodically checking whether or not the accessory 200 has a cause of the communication through polling. In addition, when the accessory 200 has the cause of the communication, the accessory 200 can communicate the fact to the camera 100 on a real-time basis.

(80) SDA connected to TA12 and SCL connected to TA13 are signals for the accessory control circuit 201 to act as a communication slave and perform the I2C communication.

(81) The FNC1 signal connected to TA14 (synchronous contact), the FNC2 signal connected to TA15, the FNC3 signal connected to TA16, and the FNC4 signal connected to TA17 are signals whose functions can be changed according to the type of the accessory 200. For example, when the accessory 200 is a microphone device, it is an audio data signal, and when the accessory 200 is a strobe device, it is a signal for controlling the light emission timing. TA14 to TA17 correspond to functional signal contacts.

(82) TA18 as the second grounded contact (reference potential contact) is also connected to GND, and serves as a reference potential contact between the camera 100 and the accessory 200, similar to TA04.

(83) The differential signal D2N connected to TA19 (first differential signal contact) and the differential signal D2P connected to TA20 (second differential signal contact) are paired data communication signals for the data communication and connected to the external connection terminal 209.

(84) TA21 is connected to GND and serves not only as a contact for a reference potential but also as a terminal for controlling the wiring impedance of the differential signals D2N and D2P. TA21 corresponds to a fourth grounded contact.

(85) The contacts TA01, TA04, TA06, TA18, and TA21 are connected, for example, to the GND portion of the flexible substrate 259 illustrated in FIGS. 19A and 19B, which will be described later, and the GND portion of the flexible substrate 259 is fixed to a metallic member having a GND level of the accessory 200 with an unillustrated screw etc. The metallic member having the GND level includes, for example, a shoe attachment leg 251 and an unillustrated base plate inside the accessory 200.

(86) FIG. 2A illustrates that the accessory connector 211 disposed on the shoe provided on the lower part of the accessory (strobe device) 200 is connected to the camera connector 141 disposed on the accessory shoe provided at the top of the camera 100. FIG. 2B illustrates an arrangement example of the 21 contacts TC01 to TC21 in the camera connector 141. TC01 is disposed at the right end viewed from the object side, and the 21 contacts up to TC21 are arranged in a row. The accessory shoe is attached by sliding it from the top side to the bottom side in FIG. 2B relative to the accessory shoe having the camera connector 141.

(87) FIG. 2C illustrates an arrangement example of the 21 contacts TA01 to TA21 in the accessory connector 211. Similar to the camera connector 141, TA01 is disposed at the right end viewed from the object side, and the 21 contacts up to TA21 are arranged in a row. Usually, the contacts TA01 to TA21 and the corresponding contacts TC01 to TC21 are connected to each other. However, if an excessive static pressure or impact is applied to the accessory 200, the contacts may be disconnected. In particular, when a force in the rotational direction acts on the direction in which the contacts are arranged in the accessory 200, the disconnection is likely to occur at the end contacts.

(88) FIG. 3A exaggerates the appearance of the excessive static pressure applied to the accessory 200 from the left side viewed from the object side. At this time, a force acts on the contacts TC21 and TA21 and the neighboring contacts of the camera connector 141 and the accessory connector 211 in the disconnecting direction, and a poor connection is likely to occur. On the other hand, a stronger force acts on the contacts TC01 and TA01 and the neighboring contacts in the connecting direction than that in the normal state.

(89) FIG. 3B exaggerates the appearance of the excessive static pressure applied to the accessory 200 from the right side viewed from the object side. At this time, a force acts on the contacts TC01 and TA01 and the neighboring contacts of the camera connector 141 and the accessory connector 211 in the disconnecting direction, and a poor connection is likely to occur. On the other hand, a stronger force acts on the contacts TC21 and TA21 and the neighboring contacts in the connecting directions than that in the normal state.

(90) This embodiment connects to GND the contacts TC01 and TA01 and TC21 and TA21 at both ends of the camera connector 141 and the accessory connector 211. Thereby, even if a poor connection temporarily occurs at the contact at one end due to the excessive static pressure, the GND connection can be secured at the contact at the other end. Therefore, this configuration can restrain each circuit and electric element from getting damaged as a result of that the reference potential of the accessory 200 becomes unstable due to the poor GND connection.

(91) When the accessory 200 in which part of the GND contacts is missing is attached due to a defect and breakdown of the accessory connector 211 or the like, the camera control circuit 101 cannot detect that the part of the GND contacts is missing. In such a case, the operating current concentrates on the remaining GND contacts, and the accessory 200 may malfunction in some

cases.

(92) FIG. 4A is a configuration example for allowing the camera **100** to detect the connection state of the GND contact of the accessory **200**, and illustrates extracted part relating to the grounded contact from the configuration illustrated in FIG. 1.

(93) TC01, TC04, TC18, and TC21 are connected to input terminals P1, P2, P3, and P4 of the camera control circuit **101**, respectively, and pulled to the camera microcomputer power supply VMCU_C via resistors **1011Rp_g1**, **1021Rp_g2**, **1031Rp_g3**, and **1041Rp_g4**, respectively. An SW circuit **1** (**1012**), an SW circuit **2** (**1022**), an SW circuit **3** (**1032**), and an SW circuit **4** (**1042**) are connected to TC01, TC04, TC18, and TC21, respectively.

(94) The SW circuit **1** is a switching circuit driven by the control signal of the camera control circuit **101**, and when it is turned on by the control signal, TC01 is connected to GND. It is desirable that the SW circuit **1** includes, for example, an FET, or a circuit that has impedance that is as small as possible when the operation is turned on, and the impedance that is as large as possible when the operation is turned off. Each of the SW circuits **2**, **3**, and **4** also has the same configuration as that of the SW circuit **1** as illustrated in FIG. 4A.

(95) A flowchart of FIG. 4B shows a sequence for determining the connection state of the grounded terminal in the configuration illustrated in FIG. 4A. The camera control circuit **101** executes this processing and other processing described later according to a computer program. S means the step.

(96) In S1001, the camera control circuit **101** monitors the signal level of the accessory attachment detection signal/ACC_DET and determines whether or not the accessory **200** is attached. If the signal level is Hi, the camera control circuit **101** returns to S1001 and performs the detection again, assuming that the accessory **200** is not attached, and if the signal level is Lo, it proceeds to S1002 assuming that the accessory **200** is attached.

(97) In S1002, the camera control circuit **101** makes such a control that the SW circuit **1** is turned on and the SW circuits **2**, **3**, and **4** are turned off, respectively. In S1003, the camera control circuit **101** confirms the voltage level of the input terminal P1, and determines that TC01 is connected to the grounded contact if it is a Lo level, and that TC01 is not connected to the grounded contact if it is a Hi level.

(98) Next, in S1004, the camera control circuit **101** makes such a control that the SW circuit **2** is turned on and the SW circuits **1**, **3**, and **4** are turned off, respectively. In S1005, the camera control circuit **101** confirms the voltage level of the input terminal P2, and determines that TC04 is connected to the grounded contact if it is a Lo level, and that TC04 is not connected to the grounded contact if it is a Hi level.

(99) Next, in S1006, the camera control circuit **101** makes such a control that the SW circuit **3** is turned on and the SW circuits **1**, **2**, and **4** are turned off, respectively.

(100) In S1007, the camera control circuit **101** confirms the voltage level of the input terminal P3, and determines that TC18 is connected to the grounded contact if it is a Lo level, and that TC18 is not connected to the grounded contact if it is a Hi level.

(101) Next, in S1008, the camera control circuit **101** makes such a control that the SW circuit **4** is turned on and the SW circuits **1**, **2**, and **3** are turned off, respectively.

(102) In S1009, the camera control circuit **101** confirms the voltage level of the input terminal P4, and determines that TC18 is connected to the grounded contact if it is a Lo level, and that it is not connected to the grounded contact if it is a Hi level.

(103) In S1010, the camera control circuit **101** makes such a control that the SW circuits **1**, **2**, **3**, and **4** are turned on, respectively.

(104) Such a control enables the camera control circuit **101** to confirm the attachment state of the grounded contact with the attached accessory **200**, and to determine whether or not to supply to the accessory power supply circuit **202** and the like based on the ground connection state.

(105) In the meanwhile, if the accessory **200** is tilted to the camera **100** or the like when the accessory **200** is attached to the camera **100**, only some of the plurality of contacts TC01 to TC21

and TA01 to TA21 can be connected with each other. As illustrated in FIGS. 16A, 16B, and 16C, where a Z direction is the attachment direction of the accessory 200 to the camera 100, an X direction is a direction in which the plurality of contacts TC01 to TC21 and TA01 to TA21 are aligned, and a Y direction is a direction orthogonal to the X direction and the Z direction, only some of the contacts may be connected in the following cases.

(106) First, as illustrated in FIGS. 3A and 3B, when the accessory 200 is tilted to the camera 100 around an axis parallel to the Z direction, some of the plurality of contacts may be connected with each other on one side where the camera 100 and the accessory 200 are close to each other but some of the plurality of contacts are disconnected from each other on the other side where the camera 100 and the accessory 200 are separated from each other. Although not illustrated, when the accessory 200 is tilted (twisted) to the camera 100 around an axis parallel to the Y direction, some contacts on an opposite side of the plurality of contacts that are connected with each other are separated from each other.

(107) As detailed later with reference to FIGS. 5A and 5B, in the camera 100 and the accessory 200 according to this embodiment, the attachment detection processing is executed prior to various communications while the accessory 200 is attached to the camera 100. At this time, if the attachment detection contacts TC06 and TA06 are connected, the attachment detection processing can be executed. After the attachment detection processing via the contacts TC06 and TA06 is executed, the communication request signal/WAKE is output from the accessory 200 to the camera 100 via the contacts (also referred to as communication request contacts hereinafter) TC11 and TA11. By detecting this communication request signal/WAKE, the camera 100 performs various communications, determining that the accessory 200 is in a communicable state. However, if the camera 100 cannot detect the communication request signal/WAKE even though the attachment of the accessory 200 to the camera 100 is detected, the camera 100 determines that there is a communication error with the accessory 200. If the accessory 200 is tilted or twisted while the accessory 200 is being attached to the camera 100, only some of the contacts are temporarily connected, it is determined that there is a communication error, and error processing, such as an alarm, is performed, the user may misunderstand that the accessory 200 breaks down.

(108) Therefore, this embodiment adopts the contact arrangement so as to reduce the occurrence of a situation in which the camera 100 cannot detect the communication request signal/WAKE even though the attachment of the accessory 200 to the camera 100 is detected.

(109) As described above, when the accessory 200 is tilted to the camera 100 around an axis parallel to the Z direction, the contacts TC01 and TA01 and the neighboring contacts are connected as illustrated in FIG. 3A, and the contacts TC21 and TA21 and the neighboring contacts are disconnected, or the contacts TC21 and TA21 and the neighboring contacts are connected and the contacts TC01 and TA01 and the neighboring contacts are disconnected as illustrated in FIG. 3B.

(110) This embodiment uses the contacts TC06 and TA06 to detect the attachment of the accessory 200 to the camera 100. As illustrated in FIG. 3A, when the contacts TC01 and TA01 are connected with each other, the neighboring contacts TC06 and TA06 are often connected with each other. At this time, if the communication request contacts TC11 and TA11 are located near the distant contacts TC21 and TA21, the camera 100 is unlikely to detect the communication request signal/WAKE even though the attachment of the accessory 200 to the camera 100 is detected.

(111) On the other hand, as illustrated in FIG. 3B, if the contacts TC06 and TA06 are connected with each other while the contacts TC21 and TA21 are connected with each other, and if the contacts TC11 and TA11 are arranged on a side of the contacts TC01 and TA01 distant from the contacts TC06 and TA06, the camera 100 is unlikely to detect the communication request signal/WAKE even though the attachment of the accessory 200 to the camera 100 is detected.

(112) On the other hand, this embodiment adopts the following contact arrangement. As illustrated in FIG. 1, the attachment detection contacts TC06 and TA06 and the communication request contacts TC11 and TA11 are disposed between the contacts TC01 and TA01 closest to one end and

the contacts **TC21** and **TA21** closest to the other end in the direction in which a plurality of contacts **TC01** to **TC21** and **TA01** to **TA21** are arranged (referred to as a contact arrangement direction hereinafter). This arrangement relationship will be referred to as a first arrangement relationship. The attachment detection contacts **TC06** and **TA06** are disposed between the communication request contacts **TC11** and **TA11** and the contacts **TC01** and **TA01**. This arrangement relationship will be referred to as a second arrangement relationship. Then, in the contact arrangement direction, distances between the attachment detection contacts **TC06** and **TA06** and the communication request contacts **TC11** and **TA11** are made shorter than distances between the communication request contacts **TC11** and **TA11** and the contacts **TC21** and **TA21**. This arrangement relationship will be referred to as a third arrangement relationship. In this embodiment, the contacts **TC01** to **TC21** and **TA01** to **TA21** are arranged at regular pitches, so that the distance between the contacts herein can be rephrased as the number of other contacts arranged between these contacts, and a short (or long) distance can be rephrased as a small (or large) number of other contacts.

(113) In this embodiment, the distances between the communication request contacts **TC11** and **TA11** and the contacts **TC01** and **TA01** are set to be equal to or less than the distance between the communication request contacts **TC11** and **TA11** and the contacts **TC21** and **TA21** in the contact arrangement direction. This arrangement relationship will be referred to as a fourth arrangement relationship. In particular, this embodiment disposes the communication request contacts **TC11** and **TA11** at the centers between the contacts **TC01** to **TC21** and **TC01** to **TC21**, and makes equal to each other the distances between the communication request contacts **TC11** and **TA11** and the contacts **TC01** and **TA01**, and the distances between the communication request contacts **TC11** and **TA11** and the contacts **TC21** and **TA21**. The communication request contacts **TC11** and **TA11** do not necessarily have to be disposed at the centers between the contacts **TC01** to **TC21** and **TC01** to **TC21**, but they are preferably disposed near the centers.

(114) This embodiment makes the distances between the attachment detection contacts **TC06** and **TA06** and the contacts **TC01** and **TA01** equal to or greater than the distances between the attachment detection contacts **TC06** and **TA06** and the communication request contacts **TC11** and **TA11** in the contact arrangement direction. This arrangement relationship will be referred to as a fifth arrangement relationship. In particular, this embodiment disposes the attachment detection contacts **TC06** and **TA06** at the centers between the communication request contacts **TC11** and **TA11** and the contacts **TC01** and **TA01**, and makes the distances between the attachment detection contacts **TC06** and **TA06** and the contacts **TC01** and **TA01** and the distances between the attachment detection contacts **TC06** and **TA06** and the communication request contacts **TC11** and **TA11** equal to each other. The attachment detection contacts **TC06** and **TA06** do not necessarily have to be disposed at the centers between the communication request contacts **TC11** and **TA11** and the contacts **TC01** and **TA01**, but it is preferable to dispose them near the centers.

(115) Due to the above contact arrangement, the communication request contacts **TC11** and **TA11** are highly likely to be connected with each other if the attachment detection contacts **TC06** and **TA06** are connected with each other in the tilted state illustrated in FIG. 3A, and in the tilted state illustrated in FIG. 3B, even if the communication request contacts **TC11** and **TA11** are connected with each other, the attachment detection contacts **TC06** and **TA06** are highly likely to be disconnected with each other. As a result, whichever state the accessory **200** is tilted in, the occurrence of a situation in which the camera **100** cannot detect the communication request signal/WAKE can be reduced even though the attachment of the accessory **200** to the camera **100** is detected.

(116) A description will now be given as a comparative example of a case where the positions of the contacts **TC06** and **TA06** and the contacts **TC11** and **TA11** are exchanged. That is, a description will be given of a case where the contacts **TC11** and **TA11** are used to detect the attachment and the contacts **TC06** and **TA06** are used to detect the communication request signal/WAKE. In this

configuration, when the accessory **200** is tilted to the camera **100** and the contacts **TC01** and **TA01** and the neighboring contacts are disconnected from each other, the contacts **TC11** and **TA11** for the attachment detection may be connected with each other but the contacts **TC06** and **TA06** for the communication request signal/WAKE may be disconnected from each other, resulting in a communication error.

(117) Therefore, in order to avoid the communication error, it is preferable to dispose the attachment detection contact on one end side in the contact arrangement direction rather than the contacts for the communication request signal/WAKE as in this embodiment.

(118) As illustrated in FIGS. **20A** to **20C** and **23**, which will be described later, in a configuration where the accessory **200** holds a plurality of contacts with a connection plug **256** as a holding member made of a nonconductive material such as a resin material, the connection plug **256** may have a convex shape toward the lower side (contact direction with the camera connector **141**) in the drawing. In such a case, the contact on one end side in the contact arrangement direction of the plurality of contacts is likely to be connected but the contact on the other end side is likely to be disconnected. However, the contact arrangement illustrated in this embodiment can reduce an occurrence of a communication error even if some of the contacts are disconnected when the accessory **200** is attached to the camera **100**.

(119) As described above, when the accessory **200** is twisted relative to the camera **100** around the axis parallel to the Y direction, some contacts on one end side in the contact arrangement direction of the plurality of contacts may be connected, but the other contacts on the other end side may be disconnected. When such a state occurs in the process of attaching the accessory **200** to the camera **100**, the connection timings shift among the plurality of contacts. If the contact timing significantly shifts, a time lag from the attachment detection of the accessory **200** to the camera **100** to the detection of WAKE becomes long, and consequently a communication error may be determined. At this time, depending on the twisting direction of the accessory **200**, the side of the contacts **TC01** and **TA01** or the side of the contacts **TC21** and **TA21** starts to be connected first.

(120) When the side of the contacts **TC01** and **TA01** starts to be connected, the closer the communication request contacts **TC11** and **TA11** are to the contacts **TC21** and **TA21**, the longer the time lag becomes from the attachment detection of the accessory **200** to the detection of the communication request signal/WAKE. The longer the time lag is, the more easily a communication error is determined. On the other hand, when the side of the contacts **TC21** and **TA21** starts to be connected and the communication request contacts **TC11** and **TA11** are disposed on the side of the contacts **TC01** and **TA01** of the attachment detection contacts **TC06** and **TA06**, a time lag occurs from the attachment detection of the accessory **200** to the detection of the communication request signal/WAKE.

(121) On the other hand, this embodiment adopts the above contact arrangement, and shortens a time lag from the attachment detection of the accessory **200** to the detection of the communication request signal/WAKE, regardless of which side of the contacts starts to be connected.

(122) This embodiment disposes at positions between the attachment detection contacts **TC06** and **TA06** and the communication request contacts **TC11** and **TA11** the contacts **TC07** and **TA07** to **TC10** and **TA10** for the SPI communication (communication in the second communication method) between the camera **100** and the accessory **200**. The contacts **TC12**, **TA12**, **TC13**, and **TA13** for the I2C communication (communication in the first communication method) between the camera **100** and the accessory **200** are disposed at positions close to the communication request contacts **TC11** and **TA11** on the opposite side of the attachment detection contacts **TC06** and **TA06**.

(123) The communication between the camera **100** and the accessory **200** is executed after the camera **100** detects the communication request signal/WAKE. Therefore, until the communication between the camera **100** and the accessory **200** is executed, the connection of the contacts for the communication is not confirmed. On the other hand, in this embodiment, if the attachment detection contacts **TC06** and **TA06** and the communication request contacts **TC11** and **TA11** are

connected with each other, the communication contacts TC07, TA07 to TC10, TA10, TC12, TA12, TC13, and TA13 arranged near and between them, respectively, can be considered to be connected. (124) Since it can be considered that the positions between the attachment detection contacts TC06 and TA06 and the communication request contacts TC11 and TA11 are more reliably connected, the contacts for the SPI communication executed after the I2C communication are preferably disposed at positions between the attachment detection contacts TC06 and TA06 and the communication request contacts TC11 and TA11.

(125) As illustrated in FIG. 4, and FIGS. 12 and 20 which will be used later for the explanation, a configuration in which the number of contacts is smaller than that of the camera 100 is conceivable as the configuration of the accessory 200. Even in this configuration, the attachment detection contact and the communication request contact are necessary contacts, and the attachment detection contact and the communication request contact may be arranged under an idea similar to the configuration in which the number of contacts is equal to that of the camera 100. However, a part of the above-mentioned first to fifth arrangement relationships may not be satisfied.

(126) For example, in the configuration having no contact TA21 as illustrated in FIG. 4, a distance between the communication request contact TA11 and the contact TA01 is longer than a distance between the communication request contact TA11 and the contact TA20 in the contact arrangement direction. That is, the fourth arrangement relationship is not satisfied. For example, in a configuration having no contacts TA01 to TA03 and TA19 to 21 as illustrated in FIG. 12, a distance between the attachment detection contact TA06 and the contact TA04 in the contact arrangement direction is shorter than a distance between the attachment detection contact TA06 and the communication request contact TA11. That is, the fifth arrangement relationship is not satisfied.

(127) As described above, in a configuration in which the position of the contact position at the end of the accessory 200 is different from the contact position at the end of the camera 100, part of the first to fifth arrangement relationships may not be satisfied. In such a case, it is assumed that the position facing the contact at the end of the camera 100 in the attached state is the position of the contact at the end of the accessory 200, and the attachment detection contact and the communication request contact may be arranged so as to satisfy the first to fifth arrangement relationships. Alternatively, as in the protrusion portion 256a illustrated in FIGS. 20A, 20B, and 20C, the attachment detection contact and the communication request contact may be arranged so as to satisfy the first to fifth arrangement relationships in consideration of the distance from the protrusion portion 256a instead of the distance from the contact at the end. A flowchart of FIG. 5A shows processing executed by the camera control circuit 101 when the accessory 200 is attached to the camera 100.

(128) In S401, the camera control circuit 101 as an attachment detection means monitors the signal level of the accessory attachment detection signal/ACC_DET and determines whether or not the accessory 200 is attached. If the signal level is Hi, the camera control circuit 101 returns to S401 and performs the detection again, assuming that the accessory 200 is not attached, and if the signal level is Lo, proceeds to S402 assuming that the accessory 200 is attached.

(129) In S402, the camera control circuit 101 sets a power supply control signal CNT_VACC1 to the Hi level in order to turn on the output of the accessory-use power supply circuit A131, and proceeds to S403. The accessory-use power supply circuit A131 outputs the accessory power supply VACC in response to the power supply control signal CNT_VACC1 becoming Hi.

(130) In S403, the camera control circuit 101 monitors the signal level of the overcurrent detection signal DET_OVC and determines whether or not the overcurrent is flowing. If the signal level is Lo, the camera control circuit 101 proceeds to S404 assuming that no overcurrent flows, and if the signal level is Hi, proceeds to S405 assuming that the overcurrent flows to perform error processing.

(131) FIG. 6A schematically illustrates a change of the signal when the flow proceeds to S404 in the processing of FIG. 5A. IACC is the current of the accessory power supply VACC. Since the

accessory power supply VACC normally leads up after the power supply control signal CNT_VACC1 is set to Hi in S402, the overcurrent detection signal DET_OVC remains at a Lo level.

(132) FIG. 6B schematically illustrates a change in the above signal when the flow proceeds to S405 in the processing of FIG. 5A. Since the overcurrent flows through IACC after the power supply control signal CNT_VACC1 is set to Hi in S402, the overcurrent detection signal DET_OVC changes to the Hi level and notifies the camera control circuit 101. Upon receiving the notification of the overcurrent detection signal DET_OVC, the camera control circuit 101 turns off the outputs of the accessory-use power supply circuits A131 and B132 as error processing to stop the power supply to the accessory 200. Thus, even when the overcurrent flows through the accessory power supply VACC, the camera control circuit 101 can detect the overcurrent and safely stop the system.

(133) Usually, in a case where an abnormal current flows through the accessory power supply VACC, it is presumed that the camera 100 and the accessory 200 are out of order, but since the camera connector 141 and the accessory connector 211 are exposed to the outside, and thus the neighboring contacts may be short-circuited due to an adhesion of a foreign matter such as a metal piece.

(134) In this embodiment, the accessory power supply VACC has a voltage of 3.3 V, whereas the camera microcomputer power supply VMCU_C and the accessory microcomputer power supply VMCU_A have a voltage of 1.8 V. Thus, if a voltage of 3.3 V is applied to an electric element operating at a voltage of 1.8 V, the electric element may get damaged. Since the post-short-circuiting behavior depends on the characteristic of the electric element, the camera control circuit 101 may not always be able to detect the short-circuiting between the terminals. For example, since the I2C communication signal is at the Hi level in the communication standby state, even if it is short-circuited with the voltage of 3.3 V equal to and higher than the voltage of 1.8V, the abnormality cannot be detected depending on the characteristic of the electric element of the connection destination.

(135) On the other hand, this embodiment disposes the GND contacts TC04 and TA04 on one of the two sides of the accessory power supply VACC contacts TC05 and TA05, and the contacts TC06 and TA06 of the accessory attachment detection signal/ACC_DET on the other side. As described above, the accessory attachment detection signal/ACC_DET is connected to GND in the accessory 200. Thus, even if short-circuiting occurs between the contacts, the overcurrent can be detected and the system can be safely stopped without applying 3.3 V to the element operating at 1.8 V.

(136) As described above, if the accessory power supply VACC is supplied when the GND contact is not connected, the reference potential of the accessory 200 becomes unstable, and consequently each circuit and the electric element may get damaged. In operating the device, an external force may be applied that makes unstable the connection of the connector terminals. On the other hand, by arranging the accessory power supply VACC contact and the GND contact adjacent to each other as in this embodiment, the connection of only the accessory power supply VACC contact can be more effectively prevented than a case where the accessory power supply VACC contact and the GND contact are separate terminals.

(137) This embodiment connects the accessory attachment detection signal/ACC_DET to GND in the accessory 200, but may connect it to GND via a resistor element Rd231 as in the accessory 200 illustrated in FIG. 9. The short-circuiting current can be reduced by the connection to GND via the resistor element Rd231.

(138) In this case, it is necessary to select the resistor element Rd231 having such a resistance value that a voltage of $(R_d/(R_p+R_d)) \times 1.8 \text{ V}$ obtained by dividing the voltage of 1.8 V of the camera microcomputer power supply VMCU_C by the resistor elements Rp134 and Rd231 satisfies the Lo level threshold (Vil) of the camera control circuit 101. For example, when a Low level detection

threshold value (Vil) of the camera control circuit **101** is 0.33 times the power supply voltage, the resistance value of the resistor element **Rd231** needs to be half or less of the resistor element **Rp134** (10 k Ω). In the example of FIG. **9**, the resistance value of the resistor element **Rd231** is set to 5 k Ω . (139) FIG. **5B** illustrates processing executed by the camera control circuit **101** when the accessory **200** having the configuration illustrated in FIG. **9** is attached to the camera **100**. Since **S411** to **S413** are the same as **S401** to **S403** illustrated in FIG. **5A**, a description thereof will be omitted.

(140) In **S414** after **S413**, the camera control circuit **101** monitors the signal level of the accessory attachment detection signal/ACC_DET, and determines whether or not the accessory attachment detection signal/ACC_DET contacts **TC06** and **TA06** are short-circuited with the accessory power supply VACC contacts **TC05** and **TA05**. If the signal level is Lo, the camera control circuit **101** proceeds to **S415** assuming that it is not short-circuited, and if the signal level is Hi, proceeds to **S416** assuming that it is short-circuited to perform the error processing.

(141) FIG. **6C** schematically illustrates the state of the above signal when the accessory power supply VACC and the accessory attachment detection signal/ACC_DET are short-circuited in the accessory **200** having the configuration of FIG. **9** to which the resistor element **Rd231** (5 k Ω) is added. After the power supply control signal CNT_VACC1 is set to Hi in **S402**, no overcurrent flows in IACC because the current is limited by the resistor element **Rd231**.

(142) On the other hand, the voltage of the accessory power supply VACC is applied to the accessory attachment detection signal/ACC_DET. As soon as the signal level of the accessory attachment detection signal/ACC_DET becomes Hi due to interrupt processing or the like, the camera control circuit **101** sets the power control signal CNT_VACC1 to Lo in the error processing and stops outputting the accessory power supply VACC (power supply to the accessory **200**). Thereby, the system can be safely stopped without continuously applying 3.3 V to the terminal of the element operating at 1.8 V.

(143) As illustrated in FIG. **10**, the accessory **200** may be controlled so that the accessory attachment detection signal/ACC_DET becomes a Lo level (GND potential) by the accessory control circuit **201** via an NPN transistor **212** as a switching means. If the accessory **200** is attached to the camera **100** in the configuration illustrated in FIG. **1**, the camera control circuit **101** can always detect the accessory **200**, but in the configuration illustrated in FIG. **10**, the accessory control circuit **201** can notify the attachment of the accessory **200** to the camera **100** at an arbitrary timing.

(144) As illustrated in FIG. **11**, the accessory **200** may be configured so as to connect the resistor element **Rd231** in series with the NPN transistor **212**. In this case, the resistance value needs to be half or less of the resistor element **Rp134** (10 k Ω) as in the configuration of FIG. **1**.

(145) As described above, even if the power supply contact and the adjacent contact are short-circuited, this embodiment can maintain the safety of the system including the camera **100** and the accessory **200**, and restrain them from getting damaged.

(146) FIG. **7** illustrates an example of the functions of the FNC1 signal to the FNC4 signal as the functional signals connected to the contacts **TC14** to **TC17** and the contacts **TA14** to **TA17** for each type of the accessory **200** (here, the microphone device and the strobe device).

(147) In the microphone device, the FNC2 signal to the FNC4 signal are used as a digital audio (I2S: Inter-IC Sound standard) data bus to transfer audio data. FIG. **8A** illustrates a configuration example of the functional circuit **206** when the accessory **200** is a microphone device.

(148) An audio processing circuit **206A1** in the functional circuit **206** is a codec circuit that converts an audio signal input from a microphone **206A2** into a digital audio (I2S) data format, and is controlled by the accessory control circuit **201**. The accessory control circuit **201** can set the sampling frequency and the resolution by controlling the audio processing circuit **206A1**. In this embodiment, the sampling frequency is 48 kHz and the resolution is 32 bits. The microphone **206A2** is, for example, a MEMS-IC microphone or an electret condenser microphone.

(149) **TA14** is the FNC1 signal that is not used as an I2S data bus and is connected to GND. In this

embodiment, the unused functional signal is connected to GND, but the disclosure is not limited to this embodiment, and a connection to a reference potential may be made as the stable potential other than the GND potential (0V) such as the power supply potential and the L level (low potential) or H level (high potential) of the signal.

(150) The FNC2 signal connected to TA15 (DATA contact) is an audio data signal (DATA), which is a signal output from the accessory **200** to the camera **100**.

(151) The FNC3 signal connected to TA16 (LRCLK contact) is an audio channel clock signal (LRCLK), which is a signal output from the accessory **200** to the camera **100**.

(152) The FNC4 signal connected to TA17 (BCLK contact) is an audio bit clock signal (BCLK), which is a signal output from the camera **100** to the accessory **200**.

(153) In this embodiment, since the sampling frequency is 48 kHz and the resolution is 32 bits as described above, the LRCLK frequency is 48 kHz and the BCLK frequency is 3.072 MHz. DATA has a maximum frequency of 1.536 MHz, which is half a cycle of CLK.

(154) In the contact arrangement according to this embodiment, the reference potential contacts TA18 and TC18 connected to the GND potential as the reference potential are disposed next to the contacts TA17 and TC17 to which the FNC4 signal (BCLK) having the highest frequency is connected among the functional signal contacts to which the functional signals are connected. The signal wiring to the accessory shoe interface is generally configured with a flexible substrate. In order to reduce the product cost, the flexible substrate may have a single-sided specification, and the substrate wiring is made in the same arrangement as the contact arrangement. This embodiment disposes the GND contact as the reference potential contact next to the functional signal contact connected to the signal having the highest frequency among the functional signals. This configuration can suppress radiation noises (EMI) from the functional signal contact, interference with signals connected to other contacts, and crosstalk with signals other than the I2S data bus.

(155) This embodiment connects to the GND potential as the reference potential the contacts TA18 and TC18 next to the contacts TA17 and TC17 connected to the FNC4 signal (BCLK) having the highest frequency, but the disclosure is not limited to this example and the same effect can be obtained even if the connection is made to a stable reference potential other than the GND potential.

(156) FIG. 8B is an example illustrating a configuration in which audio data is increased relative to FIG. 8A. The purpose of increasing the audio data is to increase the number of channels and the resolution.

(157) The FNC4 signal connected to TA17 is an audio bit clock signal (BCLK), which is the same as that illustrated in FIG. 8A.

(158) On the other hand, the FNC3 signal connected to TA14 is an audio channel clock signal (LRCLK), which is a signal output from the accessory **200** to the camera **100**.

(159) The FNC2 signal connected to TA15 is an audio data signal (DATA2), which is a signal output from the accessory **200** to the camera **100**.

(160) The FNC1 signal connected to TA16 operates as an audio data signal (DATA2), which is a signal output from the accessory **200** to the camera **100**.

(161) In this way, when an audio data signal is added to increase the audio data amount and two signals are used, arranging signals in such order that a higher frequency is disposed closer to the GND terminal can provide a configuration that is relatively highly effective in preventing crosstalk.

(162) FIG. 8C illustrates a configuration example of the functional circuit **206** when the accessory **200** is a strobe device. A light emitting circuit **206B1** in the functional circuit **206** is a strobe light emitting circuit including an IGBT, a trigger coil, and the like, and controls a light emission of the light emitter **206B2**. The light emitter **206B2** includes a xenon tube or the like and emits illumination light to illuminate the object. A charging circuit **206B3** includes a transformer, a switching FET, a capacitor, and the like, and accumulates electric charges for causing the light emitter **206B2** to emit light.

(163) The FNC1 signal connected to TA14 is a light emission synchronization signal (STARTX) for controlling the light emission timing of the light emitter 206B2, and is a signal output from the camera 100 to the accessory 200. The FNC2 signal to the FNC4 signal are not used in the strobe device, and no signal is connected to these contacts.

(164) This embodiment makes unused functional signal contacts disconnected (OPEN), but the disclosure is not limited to this embodiment, and a connection to a stable reference potential, such as the power supply potential and the L or H level of the signal, may be made according to the contacts TC15 to TC17 as connection destinations of the contacts TA15 to TA17.

(165) In the strobe device, only the FNC1 signal is used among the functional signals. Although the light emission synchronization signal (STARTX) is not a periodically generated signal, the camera 100 allocates GND to the FNC1 signal when the microphone device is connected so as to prevent the configuration of the camera control circuit 101 from being complicated.

(166) A description will now be given of further features of the contact arrangement according to this embodiment. SDA (first signal) connected to the contacts TC12 and TA12 as the first signal contacts and SCL (second signal) connected to the contacts TC13 and TA13 as the second signal contacts are both signals for the I2C communication. These signals are transmitted by the open drain communication. Since both SDA and SCL are pulled up to the camera microcomputer power supply VMCU_C, they are the signal having a relatively high impedance during the communication standby, and easily subject to crosstalk.

(167) Therefore, this embodiment assigns the communication request signal (fourth signal)/WAKE to the contacts TC11 and TA11 as the fourth signal contacts next to the SDA contacts TC12 and TA12. As described above, the communication request signal/WAKE is a signal for making a communication request from the accessory 200 to the camera 100.

(168) FIG. 15A illustrates the timing at which the accessory 200 makes a communication request to the camera 100 and performs the I2C communication. As illustrated in FIG. 15A, the signal level of the communication request signal/WAKE changes from the Hi level to the Lo level prior to the I2C communication by SCL and SDA. This is because the I2C communication is performed in response to this change. Accordingly, disposing the contacts TC11 and TA11 of the communication request signal/WAKE at and near the SDA contact for the I2C communication can keep SDA of the communication request signal/WAKE from crosstalk.

(169) As illustrated in FIG. 15A, a control to change the signal level of the communication request signal/WAKE from the Lo level to the Hi level after the I2C communication can keep SDA of the communication request signal/WAKE from crosstalk.

(170) The FNC1 signal is assigned to the contacts TC14 and TA14 as the third signal contacts next to the SCL contacts TC13 and TA13. As described above, since GND is assigned to the FNC1 signal in the microphone device, SCL can be kept from crosstalk.

(171) In the strobe device, the light emission synchronization signal (STARTX: third signal) as the FNC1 signal is assigned to the contacts TC14 and TA14 next to the SCL contacts TC13 and TA13. FIG. 15B illustrates the timing at which the accessory 200 makes a communication request to the camera 100, and performs the I2C communication and the strobe light emission. As illustrated in FIG. 15B, at the timing (period) when the light emission synchronization signal is output, no I2C communication is performed between the camera 100 and the accessory 200 in order to process the control of the strobe light emission with the highest priority. In other words, the light emission synchronization signal is a signal whose signal level changes before (or after) the I2C communication, but does not change during the I2C communication. This configuration can keep SCL of the light emission synchronization signal from crosstalk.

(172) Thus, this embodiment disposes the STARTX contact on one of both sides of the SDA contact and the SCL contact, and disposes the/WAKE contact on the other side, thereby achieving a good I2C communication.

(173) This embodiment sets to an open drain signal, similar to SDA, the communication request

signal/WAKE connected to the contacts TC11 and TA11 next to the SDA contacts TC12 and TA12. In comparison with a push-pull type communication request signal/WAKE is a push-pull system, crosstalk to SDA can be suppressed when the signal level of the communication request signal/WAKE changes.

(174) SCLK connected to the SCLK contacts TC07 and TA07 is a clock signal for the SPI communication, and operates at a driving frequency of 1 MHz in this embodiment. In this embodiment, the attachment detection contacts TC06 and TA06 next to the SCLK contacts TC07 and TA07 are used to transmit the accessory attachment detection signal/ACC_DET. As described above, the accessory attachment detection signal/ACC_DET is a signal that has a potential equivalent to GND when the accessory 200 is attached to the camera 100. Therefore, this contact arrangement can prevent crosstalk between SCLK and a signal other than the SPI bus.

(175) MOSI connected to the other contacts TC08 and TA08 next to the SCLK contacts TC07 and TA07 is a data signal transmitted from the camera control circuit 101 to the accessory control circuit 201 by the SPI communication. Generally, the timing at which the MOSI output level in the SPI communication changes is synchronized with the timing at which the SCLK output level changes. Therefore, crosstalk can be suppressed between SCLK and MOSI by disposing the MOSI contacts TC08 and TA08 next to the SCLK contacts TC07 and TA07.

(176) MISO connected to the other contacts TC09 and TA09 next to the MOSI contacts TC08 and TA08 is a data signal transmitted from the accessory control circuit 201 to the camera control circuit 101 in the SPI communication. Generally, the timing at which the MISO output level changes in the SPI communication is synchronized with the timing at which the SCLK output level changes, similar to MOSI. Therefore, by disposing the MISO contacts TC09 and TA09 next to the MOSI contacts TC08 and TA08, crosstalk between the MOSI and the MISO can be suppressed.

(177) CS connected to the other contacts TC10 and TA10 next to the MISO contacts TC09 and TA09 is a communication request signal transmitted from the camera control circuit 101 to the accessory control circuit 201 in the SPI communication. Generally, CS in the SPI communication maintains a constant output level from a communication request to a communication completion. Therefore, by disposing the CS contacts TC10 and TA10 next to the MISO contacts TC09 and TA09, crosstalk to MISO can be suppressed.

(178) The communication request signal/WAKE connected to the contacts TC11 and TA11 adjacent to the CS contacts TC10 and TA10 is a signal for making a communication request from the accessory control circuit 201 to the camera control circuit 101. As described above, the communication request signal/WAKE is an open drain signal, and thus is relatively susceptible to crosstalk. Therefore, this embodiment disposes the contacts TC10 and TA10 for CS whose signal level changes relatively infrequently, next to the contacts TC11 and TA11 for the communication request signal/WAKE, and can suppress crosstalk to the communication request signal/WAKE.

(179) A differential signal that requires an impedance control is connected to the contacts TC01 to TC03 and TA01 to TA03 and the contacts TC19 to TC21 and TA19 to TA21 located at and near both ends of the camera connector 141 and the accessory connector 211 (these will be collectively referred to as both end sides hereinafter). The signal wiring to the accessory shoe interface is generally configured with a flexible substrate. In order to achieve the desired wiring impedance on the flexible board, it is necessary to keep a distance between the lines of the differential signal and with GND to be wired in parallel. In a board in which both sides are used, it is common to form a mesh GND wiring on the backside of the differential signal. Thus, the signal wiring that requires the impedance control has relatively large restrictions on the wiring design in comparison with a general single-ended signal.

(180) On the other hand, this embodiment connects the differential signal that requires an impedance control to the contacts located on both ends of the camera connector 141 and the accessory connector 211, thereby relatively reducing a relationship with other signals and improving the degree of freedom in wiring design.

(181) The differential signal can be transferred at a high speed of about several hundred Mbps to several Gbps such as USB and PCIe, and is suitable to transfer a large data amount between devices. On the other hand, no differential signal may be used depending on the type of the accessory **200**. The accessory that uses no differential signal needs no contacts assigned to the differential signal, so the accessory cost can be reduced by deleting the contact.

(182) FIG. **12** illustrates a configuration variation of the accessory **200** illustrated in FIG. **1**. More specifically, it has such a configuration that the contacts TA**01** to TA**03** and TA**19** to TA**21** and signals and circuits connected to them are omitted. That is, the accessory **200** in FIG. **12** has 15 contacts. In the configuration in FIG. **12**, a differential signal is assigned to the contacts TC**01** to TC**03** and TC**19** to TC**21** located at both ends of the camera connector **141**. On the other hand, the accessory **200** that needs no differential signal adopts a contact arrangement that eliminates a contact for the differential signal from the accessory connector **211**, and includes only the contact necessary for the accessory **200**.

(183) The accessory **200** in FIG. **12** sets to the GND contacts the contacts TC**04** and TA**04** and the contacts TC**18** and TA**18** near both ends of the camera connector **141** and the accessory connector **211**. With such a contact arrangement, even in the accessory **200** connected to a part of the contacts of the camera connector **141**, the contacts at both ends of the accessory connector **211** can be set to the GND contacts. This configuration can prevent the GND contacts from being disconnected even when an excessive static pressure or impact is applied to the accessory **200**.

(184) A direct attachment of the accessory **200** to the camera **100** has been described. Referring now to FIG. **13**, a description will be given of an intermediate accessory **400** attached between the camera **100** and the accessory **200**. The camera **100** and the accessory **200** have the above-described configurations. The intermediate accessory **400** includes an extension cable for extending a distance between the camera **100** and the accessory **200**, an adapter that is used to simultaneously attach a plurality of accessories to the camera **100**, and the like. This embodiment will describe the intermediate accessory **400** as an extension cable. In the configuration in FIG. **13**, the intermediate accessory **400** corresponds to an accessory, and the accessory **200** corresponds to another accessory.

(185) The intermediate accessory **400** has a camera shoe and an accessory shoe that are attachable to the camera **100** and the accessory **200**, respectively, and each is provided with a camera-side intermediate connector **311** and an accessory-side intermediate connector **312**. The camera-side intermediate connector **311** has 21 contacts TM**01** to TM**21** arranged in a row, and is a connector for making an electrical connection with the camera **100**. The contacts TM**01** to TM**21** each make a one-to-one contact with the contacts TC**01** to TC**21** in the camera connector **141**.

(186) On the other hand, the accessory-side intermediate connector **312** has 21 contacts TN**01** to TN**21** arranged in a row, and is a connector for making an electrical connection with the accessory **200**. The contacts TN**01** to TN**21** each make a one-to-one contact with the contacts TA**01** to TA**21** in the accessory connector **211**.

(187) The intermediate accessory **400** having this contact arrangement can provide the power supply and the communication in the same manner as those when the accessory **200** is directly attached to the camera **100**. At this time, the intermediate accessory **400** may receive the power supply from the camera **100**, or the power supply from the camera **100** may be directly transmitted to the accessory **200**. The power supply in this embodiment includes a case where the power supply from the camera **100** is transmitted to the accessory **200** as it is and no power is supplied to the accessory **400**.

(188) In FIG. **13**, the number of contacts of the camera-side intermediate connector **311** is the same as the number of contacts of the camera connector **141**, and the number of contacts of the accessory-side intermediate connector **312** is the same as the number of contacts of the accessory connector **211**, but they may not necessarily be equal to each other.

(189) FIG. **14** illustrates a variation of the configuration of FIG. **13** about the accessory **200** and the

intermediate accessory **400**. Although differential signals are connected to the contacts TC01 to TC03 and TC19 to TC21 on both ends of the camera connector **141**, the differential signal may not be required depending on the function of the accessory **200**. The configuration of FIG. **14** eliminates the contact to which the differential signal is connected, from the camera-side intermediate connector **311** and the accessory-side intermediate connector **312** and the accessory connector **211**. That is, the intermediate accessory **400** and the accessory **200** in FIG. **14** each have 15 contacts. Thereby, a contact arrangement including only the contacts required for the intermediate accessory **400** and the accessory **200** is adopted.

(190) A detailed description will now be given of the connection configuration between the camera **100** and the external flash unit **120**, which is an example of the accessory **200**.

(191) FIG. **16A** illustrates the camera **100** viewed from the diagonally rear side. FIG. **16B** illustrates how to attach the external flash unit **120** to the accessory shoe **1123** of the camera **100**. FIG. **16C** illustrates the external flash unit **120** attached to the camera **100** viewed from the oblique rear side.

(192) The imaging optical system is provided on the front side (field side) of the camera **100**, and the image display unit **107** is provided on the rear side of the camera **100**. A top cover **150** as an exterior member is provided on the top surface of the camera **100**, and an accessory shoe **1123** is provided to the top cover **150**. On the other hand, in the external flash unit **120**, the camera connector **216** is provided at the bottom of the external flash unit **120**.

(193) As illustrated in FIG. **16B**, the external flash unit **120** is slid in a direction parallel to the front side in the Z direction (attachment side in the first direction) relative to the camera **100** to engage the camera connector **216** and the accessory shoe **1123** with each other. Thereby, the external flash unit **120** can be attached to the camera **100**. The front side in the Z direction is a direction from the rear side to the front side of the camera **100**, that is, a direction from the image display unit **107** side toward the imaging optical system side. An X direction (second direction), a Y direction (third direction), and the Z direction (front-back direction) illustrated in FIGS. **16A**, **16B**, and **16C** and subsequent figures are commonly used. The X direction is a direction orthogonal to the Z direction in the horizontal plane when the Z direction is parallel to the horizontal direction, and is the width direction of the camera **100**. The Y direction is a direction orthogonal to the Z direction and the X direction, and is the height direction of the camera **100**.

(194) A detailed description will now be given of the accessory shoe **1123** of the camera **100**. FIG. **17A** illustrates the top cover **150** and the exploded accessory shoe **1123**. FIG. **17B** illustrates the assembled accessory shoe **1123**. The assembly direction of the accessory shoe **1123** onto the top cover **150** is the Y direction.

(195) The accessory shoe **1123** includes an engagement member **151**, a connection terminal connector **152**, a shoe stage **153**, and an accessory shoe spring **154**. The engagement member **151** is a member for holding the external flash unit **120** through an engagement with the external flash unit **120**. The connection terminal connector **152** includes a plurality of connection terminals **152a** arranged at regular pitches in the X direction on a connector base member **152e** as a holding member made of a resin material or the like and held by the connector base member **152e**. The connection terminals **152a** correspond to the contacts TC01 to TC21 in the camera connector **141** illustrated in FIG. **1**.

(196) In the connection terminal connector **152**, the connection terminals **152a** are arranged on the front side in the Z direction as the attachment direction of the external flash unit **120** (on the front side of the camera **100**) as illustrated in FIG. **17B**. An engagement hole portion **156** to be engaged with a lock pin **252** of the external flash unit **120** illustrated in FIG. **19A** is provided behind the connection terminal connector **152** in the Z direction (on the rear side of the digital camera **100**).

(197) While the external flash unit **120** is attached to the accessory shoe **1123**, the connection terminals **152a** are electrically connected to the external flash unit **120**. Each of the plurality of connection terminals **152a** is electrically connected to a flexible substrate **158** disposed on the

lower side of the top cover **150** in the Y direction. The flexible substrate **158** is connected to an unillustrated main substrate of the camera **100**. Thus, when the external flash unit **120** is attached to the accessory shoe **1123**, a communication is available between the external flash unit **120** and the camera **100**.

(198) The shoe stage **153** is a housing member that encloses the engagement member **151** and the connection terminal connector **152**. An accessory shoe holding member **155** is a structural skeleton that holds the engagement member **151**. As illustrated in FIG. **17A**, the accessory shoe holding member **155**, the flexible substrate **158**, the top cover **150**, the shoe stage **153**, and the connection terminal connector **152** are fastened to the engagement member **151** by four screws **157** that are inserted into them. Thereby, these members are mutually positioned and fixed. By arranging the four screws **157** one by one in four areas equally divided in the X direction and the Z direction, the above components can be connected in a well-balanced manner.

(199) FIG. **18A** illustrates the structure on the top surface side of the engagement member **151**, and FIG. **18B** illustrates the structure on the bottom surface side of the engagement member **151**. FIG. **18C** illustrates the structure of the connection terminal connector **152** on the top surface side. FIG. **24** illustrates the accessory shoe **1123** viewed from the insertion direction of the external flash unit **120**.

(200) The engagement member **151** is formed by bending a metal plate in a loop shape so that the end surfaces of the bent ends face and contact each other at a seam **151a**. The engagement member **151** has a pair of engagement portions **151b**, and a coupler **151c** that couples the pair of engagement portions **151b** together. The engagement member **151** has a pair of first screw hole portions **151d** for fastening the screws **157** and a pair of second screw hole portions **151e**. The engagement member **151** has engagement hole portions **156** to be engaged with the lock pins **252** of the external flash unit **120**.

(201) As illustrated in FIGS. **18A** and **12**, the pair of engagement portions **151b** are separated by a first width (referred to as engagement portion interval hereinafter) **151aa** in the X direction. A holding member **254** of the external flash unit **120**, which will be described later, illustrated in FIG. **19B** is inserted into the engagement portion interval **151aa**. The pair of first screw hole portions **151d** are provided at a predetermined interval in the X direction, and serve as a pair of first fastening hole portions provided apart from each other in the X direction at the back (on the rear side) in the Z direction. The pair of second screw hole portions **151e** are provided at a predetermined interval in the X direction, and serve as a pair of second fastening hole portions provided apart from each other in the X direction at the front in the Z direction. The engagement hole portion **156** is formed at a position engageable with the lock pin **252** of the external flash unit **120** in an area sandwiched between the pair of first screw hole portions **151d**.

(202) In the connection terminal connector **152**, as illustrated in FIGS. **17B** and **18C**, a plurality of connection terminals **152a** are exposed. In the pitch direction (X direction) in which the plurality of connection terminals **152a** are aligned, the position of the camera connector **216** is determined by the engagement portion interval **151aa** of the engagement member **151**. Therefore, the holding member **254** of the external flash unit **120** is positioned relative to the connection terminal connector **152** by the engagement member **151**.

(203) A contact surface and a groove portion illustrated in FIG. **24** are formed on both sides of the connection terminal connector **152** (connector base member **152e**) as one example of the camera connector **141** illustrated in FIG. **1** sandwiching the plurality of connection terminals **152a** in the X direction on the front side in the Z direction. That is, there are formed contact surfaces **152b** that contact and position the accessory shoe **1123** in the Z direction when the external flash unit **120** is attached, and groove portions **152c** into which the accessory shoe **1123** is inserted. Each groove portion **152c** is formed so as to extend from the contact surface **152b** to the front side (attachment side) in the Z direction, and has a slope portion **152d** so as to face inwardly and diagonally upwardly (so as to have a tilt to the X direction). Part of the groove portion **152c** above the slope

portion **152d** extends outwardly in the X direction from the position of the top end of the slope portion **152d**. This is to prevent a dent (sink) from being generated in the slope portion **152d** during resin molding if the slope portion **152d** is formed up to the top end of the groove portion **152c**.

(204) As illustrated in FIG. **24**, in the X direction, an outermost inner surface **152ccc** of the groove portion **152c** in the connector base member **152e** of the accessory shoe **1123** is located outside of the inner end surfaces of the pair of engagement portions **151b** of the engagement member **151** (engagement portion interval **151aa**) and is located inside of the outermost inner surface **151bb** of the engagement member **151**.

(205) A slope start position **152cc**, which is the end (lower end) of the slope portion **152d** on the bottom surface side of the groove portion **152c**, is provided inside the engagement portion interval **151aa**. Thereby, it is possible to secure an area for providing the contact surface **152b** that comes into contact with a contact portion **251b** described later of the camera connector **216** and positions it in the Z direction. Providing the slope shape starting from the slope start position **152cc** can expand a space into which the shoe apparatus (camera connector **216** described later) of the external flash unit **120** is inserted, and can secure the degree of freedom in the shape of the shoe apparatus. As a result, the shoe apparatus of the external flash unit **120** can be sufficiently formed with a shape that protects the connection terminals.

(206) A description will now be given of the external flash unit **120**. FIG. **19A** illustrates the external flash unit **120** viewed from the camera connector **216** side (bottom side in the Y direction). FIG. **19B** is a section taken along a line A-A in FIG. **19A** and illustrates the internal structure of the camera connector **216**. FIG. **20A** illustrates the camera connector **216**. However, a base portion **250** and a lock lever **253**, which will be described later, are omitted. FIG. **20B** illustrates the camera connector **216** viewed from the front in the Z direction.

(207) The camera connector **216** is provided on the bottom side in the Y direction (top side in FIG. **19A**) of the base portion **250** of the external flash unit **120** as illustrated in FIG. **19B** when it is attached to the accessory shoe **1123** of the camera **100**. The camera connector **216** includes a shoe attachment leg (engagement member, shoe plate) **251**, the lock pins **252**, a lock lever **253**, a holding member **254**, a connection plug **256**, and a Y-direction holding member **258**.

(208) The shoe attachment leg **251** is an engagement member that engages the external flash unit **120** with the accessory shoe **1123** of the camera **100** and holds it. That is, the shoe attachment leg **251** is an engagement member on the external flash unit **120** side attachable to and detachable from the engagement member **151** of the accessory shoe **1123**.

(209) A large stress is applied to the accessory shoe **1123** and the camera connector **216** due to the pressure for maintaining the attachment state and the external force (impact, etc.) acting on the external flash unit **120**. The shoe attachment leg **251** is manufactured by processing a metal plate (sheet metal) in order to secure a high mechanical strength against such a large stress.

(210) The lock pin **252** is a member for preventing the external flash unit **120** from falling off while the camera connector **216** (shoe attachment leg **251**) is attached to the accessory shoe **1123**, and is held on the shoe attachment leg **251** movable in the Y direction. More specifically, the lock pin **252** is slidably held in the Y direction by the Y-direction holding member **258**. The lock lever **253** and the Y-direction holding member **258** are held by the holding member **254**.

(211) When the external flash unit **120** is attached to the accessory shoe **1123** and the lock lever **253** is rotated, the Y-direction holding member **258** is moved downwardly in the Y-direction in FIG. **19B** by an unillustrated cam portion. At that time, the lock pin **252** also moves downwardly in the Y direction in FIG. **19B** together with the Y-direction holding member **258**. Thereby, the lock pin **252** projects from the shoe attachment leg **251** and is engaged with the engaging hole portion **156** provided in the engagement member **151** of the accessory shoe **1123**. The lock pin **252** and the engagement hole portion **156** serve as a positioning member in the Z direction for ensuring an electrical connection between the external flash unit **120** and the camera **100**.

(212) The connection plug **256** as one example of an accessory connector **211** illustrated in FIG. **1**

is provided on the front side in the Z direction of the camera connector **216**, made of a nonconductive material (dielectric material) such as a resin material, and integrated with the holding member **254**. An outermost width T of the connection plug **256** in the X direction is narrower than a width W of the shoe attachment leg **251** in the X direction. Thereby, an area for providing the contact portion **251b** on the shoe attachment leg **251** is secured. The connection plug **256** has a plurality of connection terminals **257** for contacting and communicating with the plurality of connection terminals **152a** of the accessory shoe **1123** illustrated in FIG. **18C**. The connection terminals **257** correspond to the contacts TA01 to TA21 of the accessory connector **211** illustrated in FIG. **1**.

(213) The plurality of connection terminals **257** are provided so as to have a one-to-one correspondence with the plurality of connection terminals **152a**, and held by the holding member **254** so as to extend in the Z direction and to line up in the X direction. Each connection terminal **257** has a tip portion **257a** that comes into contact with the corresponding connection terminal **152a**. Each connection terminal **257** has a shape extending backwardly in the Z direction from a tip portion **257a**, and has an extension portion **257b** that displaces the tip portion **257a** upwardly in the Y direction in FIG. **19B** by an elastic deformation when the tip portion **257a** comes into contact with the connection terminal **152a**. A vertical extension portion **257c** extending upwardly in the Y direction is formed at the back end of the extension portion **257b** in the Z direction. Provided at the upper end of the vertical extension portion **257c** is a flexible substrate connector **257d** to be connected to an unillustrated main substrate of the external flash unit **120** and connected to a flexible substrate **259** inserted into the holding member **254** from the top side in the Y direction.

(214) The extension portion **257b** has a step portion **257e** having a step in the Y direction in the middle of the Z direction. As described above, the extension portion **257b** can be elastically deformed in the Y direction. However, when the distance L in the Z direction of the extension portion **257b** is short, a sufficient deformation amount cannot be obtained, and the durability is lowered. As a result, the connection terminal **152a** and the tip portion **257a** are repeatedly attached and detached, and the extension portion **257b** may easily get damaged. Accordingly, providing the step portion **257e** to the extension portion **257b** can secure a sufficient distance L without causing the extension portion **257b** to interfere with the shoe attachment leg **251**.

(215) As illustrated in FIGS. **20A** and **20B**, there are a pair of protrusion portions **256a** that project downwardly in the Y direction (third direction) so as to sandwich a plurality of connection terminals **257** at both ends of the connection plug **256** in the X direction. As illustrated in FIG. **20B**, a lower tip portion **256d** of each protrusion portion **256a** protrudes below a line made by connecting the lower ends of the tip portions **257a** of the connection terminals **257** in order to protect the connection terminals **257** from external forces such as the pressure and the impact. That is, the tip portions **257a** of the connection terminals **257** is provided above (inside) a line made by connecting the lower tip portions **256d** of the pair of protrusion portions **256a**.

(216) There is a slope portion **256b** on the outer side (outer surface) of each protrusion portion **256a** in the X direction, which serves as an outer surface that extends diagonally upwardly from the lower tip portion **256d** and faces diagonally downwardly, that is, has a tilt to the X direction. Since each protrusion portion **256a** has such a shape, the connection plug **256** can be inserted into the groove portion **152c** having the slope portion **152d** in the connection terminal connector **152**.

(217) The slope portion **256b** has a role of releasing an external force, such as the pressure and impact, from the connection plug **256** to prevent the connection plug from getting damaged. For example, FIG. **20C** illustrates that an external force is applied to the connection plug **256** from the X direction. FIG. **20C** illustrates the connection plug **256** viewed from the front in the Z direction.

(218) An external force from the X direction is defined as $F_{\text{sub.1}}$ as a vector. The external force $F_{\text{sub.1}}$ acting on the slope portion **256b** is decomposed according to the addition theorem in the vector space into a component force $F_{\text{sub.2}}$ in a direction along the slope portion **256b** and a component force $F_{\text{sub.3}}$ in a direction perpendicular to the slope portion **256b**. Where θ is an angle

formed by the external force $F_{sub.1}$ and the slope portion **256b**, the component force $F_{sub.2}$ and the component force $F_{sub.3}$ can be calculated by the following expression (1).

$$F_{sub.2} = F_{sub.1} \cos \theta$$

$$F_{sub.3} = F_{sub.1} \sin \theta \quad (1)$$

(219) When the slope portion **256b** is provided, θ is $0^\circ < \theta < 90^\circ$. In this range, the following is established:

$$F_{sub.2} < F_{sub.1}$$

$$F_{sub.3} < F_{sub.1} \quad (2)$$

(220) Since the component force $F_{sub.2}$ escapes in the direction along the slope portion **256b**, the component force $F_{sub.3}$ is the only force that affects the connection plug **256**. As described above, since the component force $F_{sub.3}$ is smaller than the external force $F_{sub.1}$, the connection plug **256** can be prevented from getting damaged even if an external force that is large to some extent is applied.

(221) By forming the slope portions **256b** on both sides in the X direction so that its width in the X direction becomes narrower toward the lower side in the Y direction, not only the external force from the X direction but also the external force from the bottom side in the Y direction can be partially released.

(222) FIG. **25** illustrates the partially enlarged connection plug **256** viewed from the Z direction. In the Y direction, assume that B is a height from the lower tip portion **256d** of the protrusion portion **256a** to the top surface of the connection plug **256** (a height of the connection plug including the protrusion portion), and A is a height of the slope portion **256b** from the lower tip portion **256d** (slope start position **256c**) to the upper end of the slope portion **256b**. At this time, A is preferably one-fifth or more of B, more preferably one-fourth or more, one-third or more, or half or more as illustrated in FIG. **13**. That is, the slope portion **256b** is formed to have a significant size for the function of releasing the external force from the X direction, and is different from a chamfered shape generally provided at the corner of the protrusion portion. The tilt angle θ of the slope portion **256b** to the X direction is preferably set in a range of $45^\circ \pm 20^\circ$ for the above function of releasing the external force.

(223) In order to secure a sufficient area for the contact portion **251b** on the shoe attachment leg **251** relative to the contact surface **152b** of the accessory shoe **1123** as the positioning portion in the Z direction, it is preferable to make as short as possible the width in the X direction between slope start positions **256c** at the lower tip portions **256d** of the slope portions **256b** on both sides. This embodiment sets the width between the slope start positions **256c** in the X direction inside the width V of the holding member **254** in the X direction, and thereby secures a sufficient area of the contact portion **251b**.

(224) The camera connector **216** has such a structure that the shoe attachment leg **251** and the holding member **254** are fastened. The details of this fastening structure will be described later.

(225) The holding member **254** can be inserted into the engagement portion interval **151aa** of the engagement member **151** of the accessory shoe **1123** illustrated in FIG. **18A**, and has a coupler **254a** having a width V shorter than the width W of the shoe attachment leg **251** in the X direction. The widths W and V are defined by the Japanese Industrial Standards (JIS) B7101-1975 "camera accessory attachment seat and attachment foot." When the coupler **254a** is engaged with the engagement member **151**, the position of the external flash unit **120** relative to the camera **100** is determined in the X direction. The shoe attachment leg **251** is urged upwardly in the Y direction when it comes into contact with the elastic deformer **154a** of the accessory shoe spring **154** as the urging member illustrated in FIGS. **17A** and **17B**. Thereby, the top surface of the shoe engagement portion **251a** is brought into contact (pressure contact) with the bottom surface of the engagement member **151**, and the position of the external flash unit **120** relative to the camera **100** is determined in the Y direction.

(226) When the contact portion **251b** of the shoe attachment leg **251** contacts the contact surface

152b on the front side in the Z direction of the connection terminal connector **152**, the position of the external flash unit **120** relative to the camera **100** is determined in the Z direction.

(227) The holding member **254** is also a structure for coupling the shoe attachment leg **251** and the base portion **250**, and the lock pins **252** and the connection terminals **257** are arranged inside the coupler **254a**.

(228) Next follows a description of the fastening structure between the holding member **254** and the shoe attachment leg **251**. FIG. **21A** illustrates the camera connector **216** viewed from the upper side in the Y direction, and FIG. **21B** illustrates a section taken along a line B-B in FIG. **21A**.

(229) A pair of first screw **260a** and a pair of second screws **260b**, which are fastening members for fastening the shoe attachment leg **251** to the holding member **254**, penetrate the holding member **254** and are fastened to the shoe attachment leg **251**. At this time, by disposing one screw in each of the four areas that are substantially equally divided in the X direction and the Z direction in a well-balanced manner, the shoe attachment leg **251** is stably held by the holding member **254**. As described above, the shoe attachment leg **251** is a component to which a large stress is applicable. Therefore, a required mechanical strength can be ensured by fastening the metal shoe attachment legs **251** to the holding member **254** with a pair of first screws **260a** and a pair of second screws **260b** arranged in a well-balanced manner.

(230) As illustrated in FIG. **21B**, a plurality of connection terminals **257** are arranged in an area S sandwiched by the pair of first screws **260a** and the pair of second screws **260b**. The widths between the pair of first screws **260a** and between the pair of second screws **260b** are narrower than the width between the lower tip portions **256d** of the protrusion portions **256a** of the connection plug **256**, the width V of the holding member **254**, the outermost width T of the connection plug **256**, and the width W of the shoe attachment leg **251**.

(231) FIG. **26** illustrates a section of the accessory shoe **1123** viewed from the Z direction while the camera connector **216** is attached to the accessory shoe **1123**. This figure illustrates the sizes T and V of the camera connector **216** and the positional relationship between each component of the camera connector **216** and each component of the accessory shoe **1123**.

(232) In FIG. **26**, as described above, the top surface of the shoe engagement portion **251a** of the camera connector **216** contacts the bottom (ceiling surface) of the engagement member **151** of the accessory shoe **1123** for positioning in the Y direction.

(233) On the other hand, none of the lower tip portion **256d** and the slope portion **256b** of the protrusion portion **256a** of the connection plug **256** in the camera connector **216** contact the bottom surface and the slope portion **152d** of the groove portion **152c** of the accessory shoe **1123**, respectively. A gap between the lower tip portion **256d** of the protrusion portion **256a** and the bottom surface of the groove portion **152c** of the accessory shoe **1123** is set as small as possible. Thereby, when an external force in the X direction is applied to the external flash unit **120**, the lower tip portion **256d** of the protrusion portion **256a** can come into contact with the bottom surface of the groove portion **152c** of the accessory shoe **1123**, and a floating amount of the connection plug **256** (a tilt to the accessory shoe **1123**) can be reduced.

(234) Each of a gap between the slope portions **256b** and **152d** and a gap between the inner end surface **152ccc** of the groove portion **152c** and the outer end surface of the connection plug **256** is set to be large to some extent. Thereby, when an external force in the X direction is applied to the external flash unit **120**, the connection terminals **257** and **152a** can be prevented from getting loaded.

(235) In the groove portion **152c** of the accessory shoe **1123**, a relationship between a height of the groove portion **152c** in the Y direction (a height from the bottom surface of the groove portion **152c** to a ceiling surface of the engagement member **151**) and a height of the slope portion **152d** in the Y direction is similar to a relationship between the height B of the connection plug **256** and the height A of the slope portion **256b** in the camera connector **216**. It is also preferable that the tilt angle of the slope portion **256b** to the X direction is also set in the range of $45^{\circ} \pm 20^{\circ}$, similarly to the tilt

angle θ of the slope portion **256b** in the camera connector **216**.

(236) Each embodiment described above has described a surface shape of the slope portion **256b** provided on the protrusion portion **256a** being flat, but the slope portion **256b** may be a curved surface having a curvature. That is, the slope portion **256b** may have a surface with a tilt to the X direction.

(237) This embodiment can secure an area for providing a larger number of connection terminals than ever and a shape for protecting them and an area for positioning between components, in the compact camera connector **216** and accessory shoe **1123**.

(238) A description will now be given of a variation of an external flash unit **120**. FIG. 22A illustrates the external flash unit **120** viewed from the camera connector **216** side (lower side in the Y direction). FIG. 22B illustrates a section taken along a line A-A in FIG. 22A and illustrates the internal structure of the camera connector **216**. FIG. 23A illustrates the camera connector **216**. However, the base portion **250** and the lock lever **253** are omitted. FIG. 23B illustrates the camera connector **216** viewed from the front in the Z direction.

(239) The camera connector **216** is provided on the lower side in the Y direction (upper side in FIG. 22A) of the base portion **250** of the external flash unit **120** as illustrated in FIG. 22B while it is attached to the accessory shoe **1123** of the camera **100**. The camera connector **216** has a shoe attachment leg **300a**, lock pins **252**, a lock lever **253**, a holding member **300**, a connection plug **300b**, a Y-direction holding member **258**, and a shoe cover **301**.

(240) The shoe attachment leg **300a** is an engagement member for engaging the external flash unit **120** with the accessory shoe **1123** of the camera **100**, similar to the shoe attachment leg **251** of the above-described embodiment. That is, the shoe attachment leg **300a** is an engagement member on the external flash unit **120** side attachable to and detachable from the engagement member **151** of the accessory shoe **1123**.

(241) In the above-described embodiment, the shoe attachment leg **251** as a metal shoe plate and the resin holding member **254** are formed as separate members in order to give priority to the mechanical strength. On the other hand, in this variation, the shoe attachment leg **300a** and the holding member **300** are formed as an integrated member by a resin material (nonconductive material). Thereby, the pair of first screws **260a** and the pair of second screws **260b** described in the previous embodiment are not required, a space for arranging the connection terminals **257** becomes wider, and thus a larger number of connection terminals **257** can be arranged. As a result, the external flash unit **120** can communicate more information with the camera **100** via the camera connector **216** and the accessory shoe **1123**.

(242) The connection plug **300b** is provided on the front side in the Z direction of the camera connector **216**, and formed as an integrated member with the holding member **300** made of a nonconductive resin material in this embodiment. Similar to the above-described embodiment, the outermost width T of the connection plug **300b** in the X direction is made narrower than the width W of the shoe attachment leg **300a** in the X direction, so that the area for providing the contact portion **300e** is secured in the shoe attachment leg **300a**. The connection plug **300b** has a plurality of connection terminals **257** for contacting and communicating with the plurality of connection terminals **152a** of the accessory shoe **1123** illustrated in FIG. 18C. The shoe cover **301** is an enclosure attached to the holding member **300**, and is a member that protects a plurality of connection terminals **257**. The shape of the connection terminal **257** is similar to that of the above embodiment, and the step portion **257e** is provided to secure a sufficient distance L in the Z direction of the extension portion **257b** without interfering with the shoe cover **301**.

(243) The shape of the connection plug **300b** is also similar to that of the connection plug **256** of the previous embodiment, and a pair of protrusion portions **300c** that project downwardly in the Y direction are provided so as to sandwich the plurality of connection terminals **257** at both ends of the connection plug **300b** in the X direction. As illustrated in FIG. 23B, a lower tip portion **300k** of each protrusion portion **300c** projects below a line made by connecting the lower ends of the tip

portions **257a** of the connection terminals **257** in order to protect the connection terminal **257** from the external force such as the pressure and the impact. That is, the tip portion **257a** of the connection terminal **257** is provided above (inside) a line made by connecting the lower tip portions **300k** of the pair of protrusion portions **300c**.

(244) Even in this embodiment, provided on the outer side of each protrusion portion **300c** in the X direction is a slope portion **300f** that extends diagonally upwardly from the lower tip portion **300k** and faces diagonally downwardly. Each protrusion portion **300c** having such a shape enables the connection plug **300b** to be inserted into the groove portion **152c** having the slope portion **152d** in the connection terminal connector **152** described in the previous embodiment. As described in the previous embodiment, the slope portion **300f** has a role of releasing the external force such as the pressure and the impact on the connection plug **300b** to prevent the connection plug from getting damaged.

(245) Similar to the previous embodiment, it is desirable to make as short as possible a distance in the X direction between the slope start positions **300g** at the lower tip portion **300k** of the slope portions **300f** on both sides. Therefore, the slope start positions **300g** on both sides are provided inside the width V of the holding member **254** in the X direction to sufficiently secure the area of the contact portion **300e** of the shoe attachment leg **300a**.

(246) The holding member **300** is formed so that it can be inserted into and engaged with the engagement portion interval **151aa** of the engagement member **151** illustrated in FIG. **18A**, and has a coupler **300h** having a width V shorter than the width W of the shoe attachment leg **300a** in the X direction. The width W and the width V are defined by the Japanese Industrial Standards (JIS) B7101-1975 “camera accessory attachment seat and attachment foot” as in the previous embodiment. When the coupler **300h** is engaged with the engagement member **151**, the position of the external flash unit **120** relative to the camera **100** is determined in the X direction. The shoe attachment leg **300a** is urged upwardly in the Y direction when it contacts the elastic deformer **154a** of the accessory shoe spring **154** illustrated in FIGS. **17A** and **17B**, and thereby the top surface of the shoe engagement portion **300d** contacts the bottom surface of the engagement member **151**. Thereby, the position of the external flash unit **120** relative to the camera **100** is determined in the Y direction.

(247) When the contact portion **300e** of the shoe attachment leg **300a** contacts the contact surface **152b** on the front side in the Z direction of the connection terminal connector **152**, the position of the external flash unit **120** relative to the camera **100** is determined in the Z direction. The holding member **300** is also a structure for coupling the shoe attachment legs **300a** and the base portion **250**, and the lock pin **252** and the connection terminal **257** are arranged inside the coupler **300h**.

(248) In this embodiment, the case where the camera **100**, the accessory **200**, and the intermediate accessory **400** have 21 or 15 contacts has been described, but the number of contacts may be other numbers.

(249) In this embodiment, the microphone device and the strobe device have been described as the accessory **200**, but the accessory according to the disclosure includes various devices, such as the electronic viewfinder unit, other than the microphone device and the strobe device. This embodiment has described the camera as an electronic apparatus, but the electronic apparatus according to the disclosure also includes various electronic apparatuses other than the camera.

Other Embodiments

(250) Embodiment(s) of the disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and

executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processor (CPU), microprocessor (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

(251) The disclosure can provide an electronic apparatus and an accessory, each of which can reduce the noise influence of a clock signal on a signal of an adjacent contact while suppressing an increase in the number of contacts.

(252) While the disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

Claims

1. An electronic apparatus to which an accessory is to be detachably attached, the electronic apparatus comprising a plurality of contacts electrically connectable to the accessory and arranged in a first direction orthogonal to an attachment direction of the accessory, wherein the plurality of contacts include: an attachment detection contact that is used to detect an attachment of the accessory to the electronic apparatus; a first clock contact that is used to communicate a first clock signal in communication between the electronic apparatus and the accessory; and a second clock contact that is used to communicate a second clock signal in the communication between the electronic apparatus and the accessory, wherein a potential of the attachment detection contact changes from a predetermined potential to a reference potential different from the predetermined potential after the accessory is attached to the electronic apparatus, wherein a frequency of a change in a potential of the first clock signal is higher than a frequency of a change in a potential of the second clock signal, and wherein the attachment detection contact is disposed next to the first clock contact and is not disposed next to the second clock contact.
2. The electronic apparatus according to claim 1, wherein the plurality of contacts further include a first differential signal contact and a second differential signal contact that are adjacent to each other and used to communicate a pair of differential signals, which are third clock signals or data signals, in the communication between the electronic apparatus and the accessory, and wherein the attachment detection contact is disposed at a position that is not next to the first or second differential signal contact.
3. The electronic apparatus according to claim 1, wherein the plurality of contacts further include a first data contact that is used to communicate a first data signal to be transmitted in communication using the first clock signal between the electronic apparatus and the accessory, and wherein the first data contact is disposed next to the first clock contact and on a side opposite to the attachment detection contact with respect to the first clock contact.
4. The electronic apparatus according to claim 3, wherein the plurality of contacts further include a second data contact that is used to communicate a second data signal received in communication using the first clock signal between the electronic apparatus and the accessory, a third data contact having a frequency of a change in a potential of a signal is lower than that of each of the first and second data contacts in the communication using the first clock signal between the electronic

apparatus and the accessory, and a communication request contact that is used for a communication request from the accessory to the electronic apparatus, and wherein the third data contact is disposed next to the communication request contact and more distant from the first clock contact than the first and second data contacts.

5. The electronic apparatus according to claim 1, wherein the plurality of contacts further include a power supply contact, and a reference potential contact that is connected to a reference potential, each of which is disposed on a side opposite to the first clock contact with respect to the attachment detection contact.

6. An accessory detachably attached to an electronic apparatus and comprising a plurality of contacts electrically connectable to the electronic apparatus and arranged in a first direction orthogonal to an attachment direction to the electronic apparatus, wherein the plurality of contacts include: an attachment detection contact that is used to detect an attachment of the accessory to the electronic apparatus; a first clock contact that is used to communicate a first clock signal in communication between the electronic apparatus and the accessory; and a second clock contact that is used to communicate a second clock signal in the communication between the electronic apparatus and the accessory, wherein the attachment detection contact is connected to a reference potential, wherein a frequency of a change in a potential of the first clock signal is higher than a frequency of a change in a potential of the second clock signal, and wherein the attachment detection contact is disposed next to the first clock contact and is not disposed next to the second clock contact.

7. The accessory according to claim 6, wherein the plurality of contacts further include a first differential signal contact and a second differential signal contact that are adjacent to each other and used to communicate a pair of differential signals, which are third clock signals or data signals, in the communication between the electronic apparatus and the accessory, and wherein the attachment detection contact is disposed at a position that is not next to the first or second differential signal contact.

8. The accessory according to claim 6, wherein the plurality of contacts further include a first data contact that is used to communicate a first data signal to be transmitted in communication using the first clock signal between the electronic apparatus and the accessory, and wherein the first data contact is disposed next to the first clock contact and on a side opposite to the attachment detection contact with respect to the first clock contact.

9. The accessory according to claim 8, wherein the plurality of contacts further include a second data contact that is used to communicate a second data signal to be transmitted in the communication using the first clock signal between the electronic apparatus and the accessory, a third data contact having a frequency of a change in a potential of a signal is lower than that of each of the first and second data contacts in the communication using the first clock signal between the electronic apparatus and the accessory, and a communication request contact that is used for a communication request from the accessory to the electronic apparatus, and wherein the third data contact is disposed next to the communication request contact and more distant from the first clock contact than the first and second data contacts.

10. The accessory according to claim 6, wherein the plurality of contacts further include a power supply contact that is used to supply power from the electronic apparatus to the accessory, and a reference potential contact that is connected to a reference potential, each of which is disposed on a side opposite to the first clock contact with respect to the attachment detection contact.

11. The accessory according to claim 6, wherein the accessory comprises a connector having protrusions protruding in a direction orthogonal to the attachment direction to the electronic apparatus and the first direction at positions on both outer sides of the plurality of contacts in the first direction.

12. The accessory according to claim 11, wherein each protrusion includes a slope portion on a side not facing the plurality of contacts so that a width of the protrusion in the first direction located at a

tip position in a protruding direction is smaller than that located at a position distant from a tip in the protruding direction.

13. The accessory according to claim 6, wherein the accessory is an intermediate accessory attachable between the electronic apparatus and another accessory.

14. An accessory comprising a contact that is used for communication with the electronic apparatus via an intermediate accessory that is the accessory according to claim 13.

15. An electronic apparatus to which an accessory is to be detachably attached, the electronic apparatus comprising a plurality of contacts electrically connectable to the accessory and arranged in a first direction orthogonal to an attachment direction of the accessory, wherein the plurality of contacts include: an attachment detection contact that is used to detect an attachment of the accessory to the electronic apparatus; a first clock contact that is used to communicate a first clock signal in communication between the electronic apparatus and the accessory; a first data contact that is used to communicate a first data signal to be transmitted in communication using the first clock signal between the electronic apparatus and the accessory; a second data contact that is used to communicate a second data signal received in communication using the first clock signal between the electronic apparatus and the accessory; a third data contact having a frequency of a change in a potential of a signal is lower than that of each of the first and second data contacts in the communication using the first clock signal between the electronic apparatus and the accessory; and a communication request contact that is used for a communication request from the accessory to the electronic apparatus, wherein a potential of the attachment detection contact changes from a predetermined potential to a reference potential different from the predetermined potential after the accessory is attached to the electronic apparatus, wherein the attachment detection contact is disposed next to the first clock contact, wherein the first data contact is disposed next to the first clock contact and on a side opposite to the attachment detection contact with respect to the first clock contact, and wherein the third data contact is disposed next to the communication request contact and more distant from the first clock contact than the first and second data contacts.

16. An accessory detachably attached to an electronic apparatus and comprising a plurality of contacts electrically connectable to the electronic apparatus and arranged in a first direction orthogonal to an attachment direction to the electronic apparatus, wherein the plurality of contacts include: an attachment detection contact that is used to detect an attachment of the accessory to the electronic apparatus; a first clock contact that is used to communicate a first clock signal in communication between the electronic apparatus and the accessory; a first data contact that is used to communicate a first data signal to be transmitted in communication using the first clock signal between the electronic apparatus and the accessory; a second data contact that is used to communicate a second data signal to be transmitted in the communication using the first clock signal between the electronic apparatus and the accessory; a third data contact having a frequency of a change in a potential of a signal is lower than that of each of the first and second data contacts in the communication using the first clock signal between the electronic apparatus and the accessory; and a communication request contact that is used for a communication request from the accessory to the electronic apparatus, wherein the attachment detection contact is connected to a reference potential, and wherein the attachment detection contact is disposed next to the first clock contact, wherein the first data contact is disposed next to the first clock contact and on a side opposite to the attachment detection contact with respect to the first clock contact, and wherein the third data contact is disposed next to the communication request contact and more distant from the first clock contact than the first and second data contacts.
